

# Chapter 1

## Introduction

### 1.1 Graphs and Graph Models

A major publishing company has ten editors (referred to by  $1, 2, \dots, 10$ ) in the scientific, technical, and computing areas. These ten editors have a standard meeting time during the first Friday of every month and have divided themselves into seven committees to meet later in the day to discuss specific topics of interest to the company, namely, advertising, securing reviewers, contacting new potential authors, finances, used copies and new editions, competing textbooks, and textbook representatives. This leads us to our first example.

**Example 1.1** The ten editors have decided on the seven committees:  $c_1 = \{1, 2, 3\}$ ,  $c_2 = \{1, 3, 4, 5\}$ ,  $c_3 = \{2, 5, 6, 7\}$ ,  $c_4 = \{4, 7, 8, 9\}$ ,  $c_5 = \{2, 6, 7\}$ ,  $c_6 = \{8, 9, 10\}$ ,  $c_7 = \{1, 3, 9, 10\}$ . They have set aside three time periods for the seven committees to meet on those Fridays when all ten editors are present. Some pairs of committees cannot meet during the same period because one or two of the editors are on both committees. This situation can be modeled visually as shown in Figure 1.1.

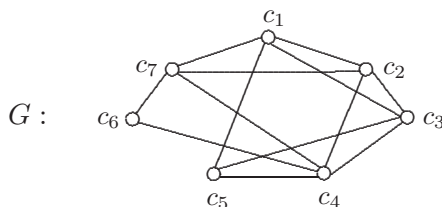


Figure 1.1: A graph

In this figure, there are seven small circles, representing the seven committees and a straight line segment is drawn between two circles if the committees they represent have at least one committee member in common. In other words, a straight line segment between two small circles (committees) tells us that these two committees should not be scheduled to meet at the same time. This gives us a picture or a “model” of the committees and the overlapping nature of their membership.  $\diamond$

What we have drawn in Figure 1.1 is called a graph. Formally, a **graph**  $G$  consists of a finite nonempty set  $V$  of objects called **vertices** (the singular is **vertex**) and a set  $E$  of 2-element subsets of  $V$  called **edges**. The sets  $V$  and  $E$  are the **vertex set** and **edge set** of  $G$ , respectively. So a graph  $G$  is a pair (actually an *ordered* pair) of two sets  $V$  and  $E$ . For this reason, some write  $G = (V, E)$ . At times, it is useful to write  $V(G)$  and  $E(G)$  rather than  $V$  and  $E$  to emphasize that these are the vertex and edge sets of a particular graph  $G$ . Although  $G$  is the common symbol to use for a graph, we also use  $F$  and  $H$  as well as  $G'$ ,  $G''$  and  $G_1$ ,  $G_2$ , etc. Vertices are sometimes called **points** or **nodes** and edges are sometimes called **lines**. Indeed, there are some who use the term **simple graph** for what we call a graph. Two graphs  $G$  and  $H$  are **equal** if  $V(G) = V(H)$  and  $E(G) = E(H)$ , in which case we write  $G = H$ .

It is common to represent a graph by a diagram in the plane (as we did in Figure 1.1) where the vertices are represented by points (actually small circles – open or solid) and whose edges are indicated by the presence of a line segment or curve between the two points in the plane corresponding to the appropriate vertices. The diagram itself is then referred to as a graph. For the graph  $G$  of Figure 1.1 then, the vertex set of  $G$  is  $V(G) = \{c_1, c_2, \dots, c_7\}$  and the edge set of  $G$  is

$$\begin{aligned} E(G) = & \{ \{c_1, c_2\}, \{c_1, c_3\}, \{c_1, c_5\}, \{c_1, c_7\}, \{c_2, c_3\}, \{c_2, c_4\}, \{c_2, c_7\}, \\ & \{c_3, c_4\}, \{c_3, c_5\}, \{c_4, c_5\}, \{c_4, c_6\}, \{c_4, c_7\}, \{c_6, c_7\} \}. \end{aligned}$$

Let's consider another situation. Have you ever encountered this sequence of integers before?

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \dots$$

Every integer in the sequence is the sum of the two integers immediately preceding it (except for the first two integers of course). These numbers are well-known in mathematics and are called the **Fibonacci numbers**. In fact, these integers occur so often that there is a journal (*The Fibonacci Quarterly*, frequently published *five* times a year!) devoted to the study of their properties. Our second example concerns these numbers.

**Example 1.2** Consider the set  $S = \{2, 3, 5, 8, 13, 21\}$  of six specific Fibonacci numbers. There are some pairs of distinct integers belonging to  $S$  whose sum or difference (in absolute value) also belongs to  $S$ , namely,  $\{2, 3\}$ ,  $\{2, 5\}$ ,  $\{3, 5\}$ ,  $\{3, 8\}$ ,  $\{5, 8\}$ ,  $\{5, 13\}$ ,  $\{8, 13\}$ ,  $\{8, 21\}$ , and  $\{13, 21\}$ . There is a more visual way

of identifying these pairs, namely, by the graph  $H$  of Figure 1.2. In this case,  $V(H) = \{2, 3, 5, 8, 13, 21\}$  and

$$E(H) = \{\{2, 3\}, \{2, 5\}, \{3, 5\}, \{3, 8\}, \{5, 8\}, \{5, 13\}, \{8, 13\}, \{8, 21\}, \{13, 21\}\}. \diamond$$

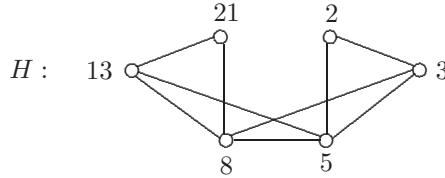


Figure 1.2: Another graph

When dealing with graphs, it is customary and simpler to write a 2-element set  $\{u, v\}$  as  $uv$  (or  $vu$ ). If  $uv$  is an edge of  $G$ , then  $u$  and  $v$  are said to be **adjacent** in  $G$ . The number of vertices in  $G$  is often called the **order** of  $G$ , while the number of edges is its **size**. Since the vertex set of every graph is nonempty, the order of every graph is at least 1. A graph with exactly one vertex is called a **trivial graph**, implying that the order of a **nontrivial graph** is at least 2. The graph  $G$  of Figure 1.1 has order 7 and size 13, while the graph  $H$  of Figure 1.2 has order 6 and size 9. We often use  $n$  and  $m$  for the order and size, respectively, of a graph. So, for the graph  $G$  of Figure 1.1,  $n = 7$  and  $m = 13$ ; while for the graph  $H$  of Figure 1.2,  $n = 6$  and  $m = 9$ .

A graph  $G$  with  $V(G) = \{u, v, w, x, y\}$  and  $E(G) = \{uv, uw, vw, vx, wx, xy\}$  is shown in Figure 1.3(a). There are occasions when we are interested in the structure of a graph and not in what the vertices are called. In this case, a graph is drawn without labeling its vertices. For this reason, the graph  $G$  of Figure 1.3(a) is a **labeled graph** and Figure 1.3(b) represents an **unlabeled graph**.

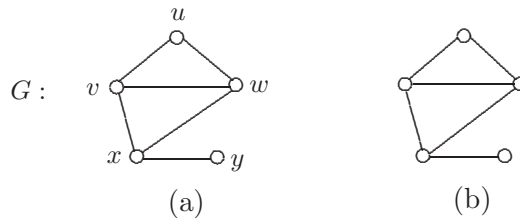


Figure 1.3: A labeled graph and an unlabeled graph

Let us now turn to yet another situation.

**Example 1.3** Suppose that we have two coins, one silver and one gold, placed on two of the four squares of a  $2 \times 2$  checkerboard. There are twelve such configurations, shown in Figure 1.4, where the shaded coin is the gold coin.

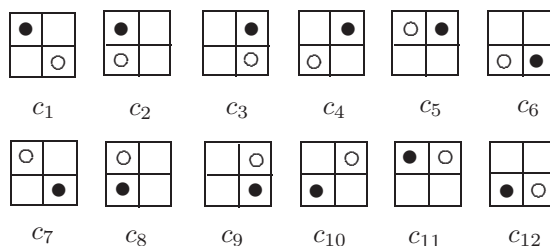
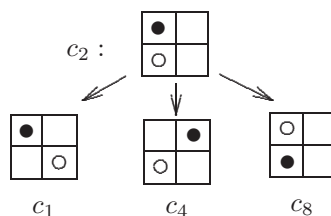


Figure 1.4: Twelve configurations

A configuration can be transformed into other configurations according to certain rules. Specifically, we say that the configuration  $c_i$  can be transformed into a configuration  $c_j$  ( $1 \leq i, j \leq 12, i \neq j$ ) if  $c_j$  can be obtained from  $c_i$  by performing exactly one of the following two steps:

- (1) moving one of the coins in  $c_i$  horizontally or vertically to an unoccupied square;
- (2) interchanging the two coins in  $c_i$ .

Necessarily, if  $c_i$  can be transformed into  $c_j$ , then  $c_j$  can be transformed into  $c_i$ . For example,  $c_2$  can be transformed into (i)  $c_1$  by shifting the silver coin in  $c_2$  to the right, (ii)  $c_4$  by shifting the gold coin to the right, or (iii)  $c_8$  by interchanging the two coins (see Figure 1.5).

Figure 1.5: Transformations of the configuration  $c_2$ 

Now consider the twelve configurations shown in Figure 1.4. Some pairs  $c_i, c_j$  of these configurations, where  $1 \leq i, j \leq 12, i \neq j$ , can be transformed into each other, and some pairs cannot. This situation can also be represented by a graph, say by a graph  $F$  where  $V(F) = \{c_1, c_2, \dots, c_{12}\}$  and  $c_i c_j$  is an edge of  $F$  if  $c_i$  and  $c_j$  can be transformed into each other. This graph  $F$  is shown in Figure 1.6.  $\diamond$

Let's look at a somewhat related example.

**Example 1.4.** Suppose that we have a collection of 3-letter English words, say

ACT, AIM, ARC, ARM, ART, CAR, CAT, OAR, OAT, RAT, TAR.

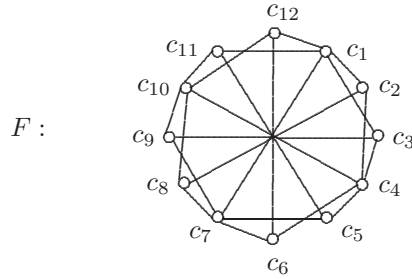


Figure 1.6: Modeling transformations of twelve configurations

We say that a word  $W_1$  can be transformed into a word  $W_2$  if  $W_2$  can be obtained from  $W_1$  by performing exactly one of the following two steps:

- (1) interchanging two letters of  $W_1$ ;
- (2) replacing a letter in  $W_1$  by another letter.

Therefore, if  $W_1$  can be transformed into  $W_2$ , then  $W_2$  can be transformed into  $W_1$ . This situation can be modeled by a graph  $G$ , where the given words are the vertices of  $G$  and two vertices are adjacent in  $G$  if the corresponding words can be transformed into each other. This graph is called **the word graph of the set of words**. For the 11 words above, its word graph  $G$  is shown in Figure 1.7.

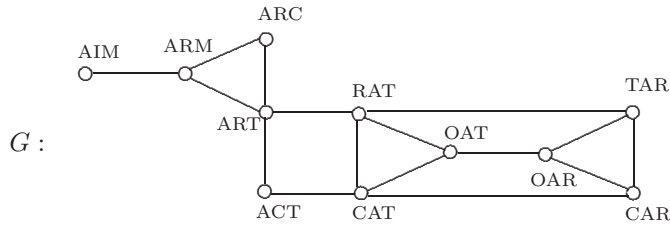


Figure 1.7: The word graph of a set of 11 words

In this case, a graph  $G$  is called **a word graph** if  $G$  is the word graph of some set  $S$  of 3-letter words. For example, the (unlabeled) graph  $G$  of Figure 1.8(a) is a word graph because it is the word graph of the set  $S = \{\text{BAT}, \text{BIT}, \text{BUT}, \text{BAD}, \text{BAR}, \text{CAT}, \text{HAT}\}$ , as shown in Figure 1.8(b). (This idea is related to the concept of “isomorphic graphs”, which will be discussed in detail in Chapter 3.)  $\diamond$

We conclude this section with one last example.

**Example 1.5** Figure 1.9 shows the traffic lanes at the intersection of two busy streets. When a vehicle approaches this intersection, it could be in one of the nine lanes: L1, L2, ..., L9.

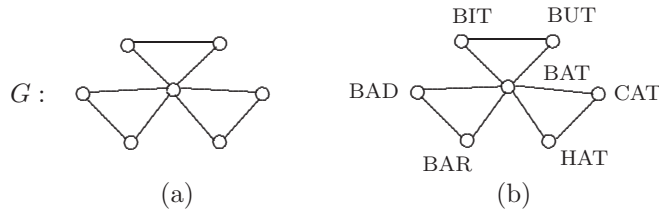


Figure 1.8: A word graph

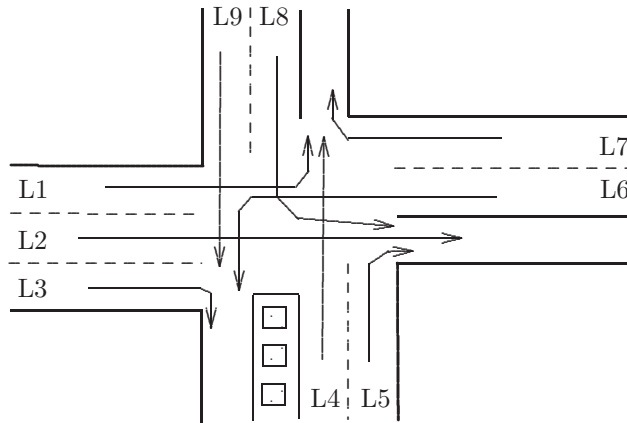
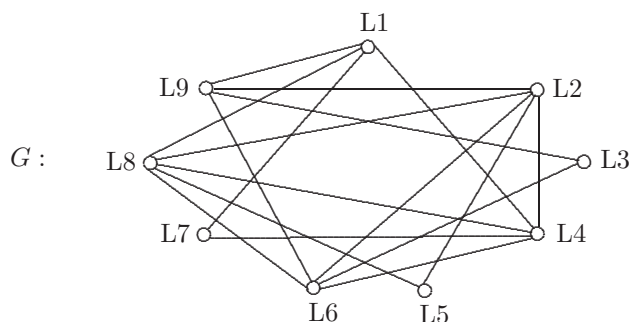


Figure 1.9: Traffic lanes at street intersections

This intersection has a traffic light that informs drivers in vehicles in the various lanes when they are permitted to proceed through the intersection. To be sure, there are pairs of lanes containing vehicles that should not enter the intersection at the same time, such as L1 and L7. However, there would be no difficulty for vehicles in L1 and L5 to drive through this intersection at the same time. This situation can be represented by the graph  $G$  of Figure 1.10, where  $V(G) = \{L1, L2, \dots, L9\}$  and two vertices (lanes) are joined by an edge if vehicles in these two lanes cannot safely enter the intersection at the same time, as there would be a possibility of an accident.  $\diamond$

What we have just seen is how five different situations can be represented by graphs. Actually, in each case, there is a set involved: (1) a set of committees; (2) a set of integers, (3) a set of configurations consisting of two coins on a  $2 \times 2$  checkerboard, (4) a set of 3-letter words, and (5) a set of traffic lanes at a street intersection. Certain pairs of elements in each set are related in some manner: (1) two committees have a member in common; (2) the sum or difference (in absolute value) of two integers in the set also belongs to the set; (3) two configurations can be transformed into each other according to some rule; (4) two 3-letter words can be transformed into each other by certain movements of letters; and (5) cars

Figure 1.10: The graph  $G$  in Example 1.5

in certain pairs of traffic lanes cannot enter the intersection at the same time. In each case, a graph  $G$  is defined whose vertices are the elements of the set and two vertices of  $G$  are adjacent if they are related as described above. The graph  $G$  then *models* the given situation. Often questions concerning the situations described above arise and can be analyzed by studying the graphs that model them.

### Exercises for Section 1.1

- 1.1 What is a logical question to ask in Example 1.1? Answer this question.
- 1.2 Create an example of your own similar to Example 1.1 with nine editors and eight committees and then draw the corresponding graph.
- 1.3 Let  $S = \{2, 3, 4, 7, 11, 13\}$ . Draw the graph  $G$  whose vertex set is  $S$  and such that  $ij \in E(G)$  for  $i, j \in S$  if  $i + j \in S$  or  $|i - j| \in S$ .
- 1.4 Let  $S = \{-6, -3, 0, 3, 6\}$ . Draw the graph  $G$  whose vertex set is  $S$  and such that  $ij \in E(G)$  for  $i, j \in S$  if  $i + j \in S$  or  $|i - j| \in S$ .
- 1.5 Create your own set  $S$  of integers and draw the graph  $G$  whose vertex set is  $S$  and such that  $ij \in E(G)$  if  $i$  and  $j$  are related by some rule imposed on  $i$  and  $j$ .
- 1.6 Consider the twelve configurations  $c_1, c_2, \dots, c_{12}$  in Figure 1.4. For every two configurations  $c_i$  and  $c_j$ , where  $1 \leq i, j \leq 12$ ,  $i \neq j$ , it may be possible to obtain  $c_j$  from  $c_i$  by first shifting one of the coins in  $c_i$  horizontally or vertically *and* then interchanging the two coins. Model this by a graph  $F$  such that  $V(F) = \{c_1, c_2, \dots, c_{12}\}$  and  $c_i c_j$  is an edge of  $F$  if  $c_i$  and  $c_j$  can be transformed into each other by this 2-step process.

1.7 Following Example 1.4,

- (a) give an example of ten 3-letter words, none of which are mentioned in Example 1.4, and whose corresponding word graph has at least six edges. Draw this graph.
- (b) give a set of five 3-letter words whose word graph is shown in Figure 1.11 (with the vertices appropriately labeled).

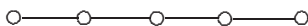


Figure 1.11: The graph in Exercise 1.7(b)

- (c) give a set of five 3-letter words whose word graph is shown in Figure 1.12 (with the vertices appropriately labeled).

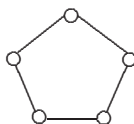


Figure 1.12: The graph in Exercise 1.7(c)

1.8 Let  $S$  be a finite set of 3-letter and/or 4-letter words. In this case, the word graph  $G(S)$  of  $S$  is that graph whose vertex set is  $S$  and such that two vertices (words)  $w_1$  and  $w_2$  are adjacent if either (1) or (2) below occurs:

- (1) one of the words can be obtained from the other by replacing one letter by another letter,
  - (2)  $w_1$  is a 3-letter word and  $w_2$  is a 4-letter word, and  $w_2$  can be obtained from  $w_1$  by the insertion of a single letter (anywhere, including the beginning or the end) into  $w_1$ .
- (a) Find six sets  $S_1, S_2, \dots, S_6$  of 3-letter and/or 4-letter words so that for each integer  $i$  ( $1 \leq i \leq 6$ ) the graph  $G_i$  of Figure 1.13 is the word graph of  $S_i$ .
  - (b) For another graph  $H$  (of your choice), determine whether  $H$  is a word graph of some set.

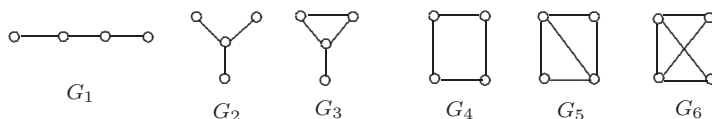


Figure 1.13: The graphs for Exercise 1.8(a)



1.9 Define a word graph differently from the word graphs defined in Example 1.4 and Exercise 1.8 and illustrate your definition.

1.10 Figure 1.14 illustrates the traffic lanes at the intersection of two streets. When a vehicle approaches this intersection, it could be in one of the seven lanes: L1, L2, ..., L7. Draw a graph  $G$  that models this situation, where  $V(G) = \{L1, L2, \dots, L7\}$  and where two vertices are joined by an edge if vehicles in these two lanes cannot safely enter this intersection at the same time.

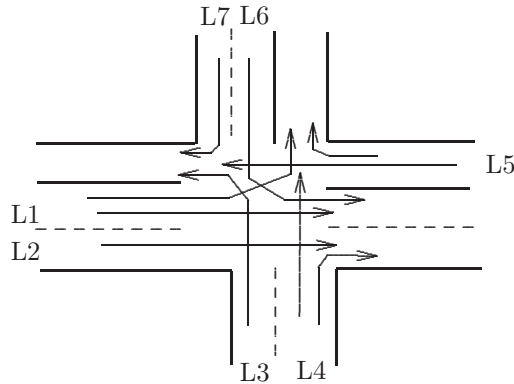


Figure 1.14: Traffic lanes at a street intersection in Exercise 1.10

## 1.2 Connected Graphs

In order to analyze certain situations that can be modeled by graphs, we must have a better understanding of graphs. As with all areas of mathematics, there is a certain amount of terminology with which we must be familiar in order to discuss graphs and their properties. Learning this fundamental terminology is our current goal. First, let's review some concepts and introduce others. A **graph**  $G$  consists of a finite nonempty set  $V$  of **vertices** and a set  $E$  of 2-element subsets of  $V$  called **edges**. If  $e = uv$  is an edge of  $G$ , then  $u$  and  $v$  are **adjacent vertices**. We also say  $u$  and  $v$  are **joined** by the edge  $e$ . The vertices  $u$  and  $v$  are referred to as **neighbors** of each other. In this case, the vertex  $u$  and the edge  $e$  (as well as  $v$  and  $e$ ) are said to be **incident** with each other. Distinct edges incident with a common vertex are **adjacent edges**.

As we mentioned earlier, although graphs are defined in terms of sets, it is customary and convenient to represent graphs by (and, in fact, to consider them as) diagrams. A graph  $G$  with vertex set  $V = \{u, v, w, x, y\}$  and edge set  $E = \{uv, vw, vx, vy, wy, xy\}$  is shown in Figure 1.15. Since this graph has five vertices and six edges, its order is 5 and its size is 6. In this graph  $G$ , the

vertices  $u$  and  $v$  are adjacent, while  $u$  and  $w$  are not adjacent. The vertex  $v$  is incident with the edge  $vw$  but not with the edge  $wy$ . The edges  $uv$  and  $vw$  are adjacent, but  $uv$  and  $xy$  are not adjacent.

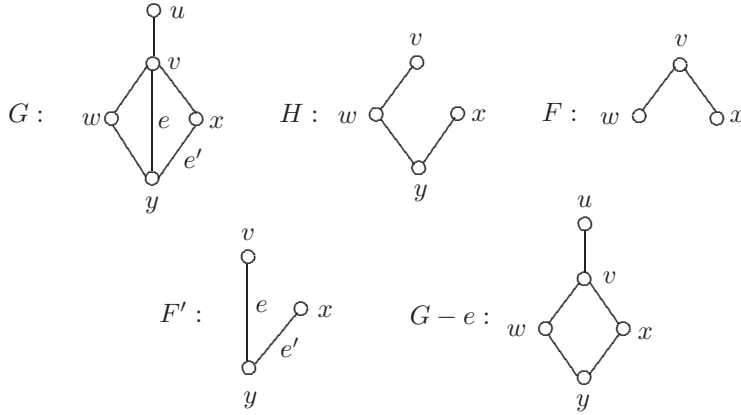


Figure 1.15: A graph  $G$  and some of its subgraphs

A graph  $H$  is called a **subgraph** of a graph  $G$ , written  $H \subseteq G$ , if  $V(H) \subseteq V(G)$  and  $E(H) \subseteq E(G)$ . We also say that  $G$  **contains**  $H$  as a subgraph. If  $H \subseteq G$  and either  $V(H)$  is a proper subset of  $V(G)$  or  $E(H)$  is a proper subset of  $E(G)$ , then  $H$  is a **proper subgraph** of  $G$ . So the graph  $H$  of Figure 1.15 is a subgraph of the graph  $G$  shown in that figure; indeed,  $H$  is a proper subgraph of  $G$ . If a subgraph of a graph  $G$  has the same vertex set as  $G$ , then it is a **spanning subgraph** of  $G$ .

A subgraph  $F$  of a graph  $G$  is called an **induced subgraph** of  $G$  if whenever  $u$  and  $v$  are vertices of  $F$  and  $uv$  is an edge of  $G$ , then  $uv$  is an edge of  $F$  as well. Therefore, the graph  $H$  of Figure 1.15 is not an induced subgraph of  $G$  of Figure 1.15 since, for example,  $v, x \in V(H)$  and  $vx \in E(G)$ , but  $vx \notin E(H)$ . On the other hand, the graph  $F$  of Figure 1.15 is an induced subgraph of  $G$ . If  $S$  is a nonempty set of vertices of a graph  $G$ , then the **subgraph of  $G$  induced by  $S$**  is the induced subgraph with vertex set  $S$ . This induced subgraph is denoted by  $\langle S \rangle$ . To emphasize that this is an induced subgraph of  $G$ , we sometimes denote this subgraph by  $\langle S \rangle_G$ . For a nonempty set  $X$  of edges, the **subgraph  $\langle X \rangle$  induced by  $X$**  has edge set  $X$  and consists of all vertices that are incident with at least one edge in  $X$ . This subgraph is called an **edge-induced subgraph** of  $G$ . Sometimes  $G[S]$  and  $G[X]$  are used for  $\langle S \rangle$  and  $\langle X \rangle$ , respectively. The graph  $F'$  of Figure 1.15 is an edge-induced subgraph of  $G$  of that figure; indeed,  $F' = \langle X' \rangle$ , where  $X' = \{e, e'\}$ .

Any proper subgraph of a graph  $G$  can be obtained by removing vertices and edges from  $G$ . For an edge  $e$  of  $G$ , we write  $G - e$  for the spanning subgraph of  $G$  whose edge set consists of all edges of  $G$  except  $e$ . More generally, if  $X$  is a set of edges of  $G$ , then  $G - X$  is the spanning subgraph of  $G$  with  $E(G - X) = E(G) - X$ .

For the graph  $G$  of Figure 1.15 and  $e = vy$ , the subgraph  $G - e$  is shown. If  $X = \{e_1, e_2, \dots, e_k\}$ , then we also write  $G - X$  as  $G - e_1 - e_2 - \dots - e_k$ .

For a vertex  $v$  of a nontrivial graph  $G$ , the subgraph  $G - v$  consists of all vertices of  $G$  except  $v$  and all edges of  $G$  except those incident with  $v$ . For a proper subset  $U$  of  $V(G)$ , the subgraph  $G - U$  has vertex set  $V(G) - U$  and its edge set consists of all edges of  $G$  joining two vertices in  $V(G) - U$ . Necessarily,  $G - U$  is an induced subgraph of  $G$ ; indeed,  $G - U = \langle V(G) - U \rangle$ . For  $U = \{u, y\}$  in the graph  $G$  of Figure 1.15,  $G - U$  is the subgraph  $F$  shown in that figure.

If  $u$  and  $v$  are nonadjacent vertices of a graph  $G$ , then  $e = uv \notin E(G)$ . By  $G + e$ , we mean the graph with vertex set  $V(G)$  and edge set  $E(G) \cup \{e\}$ . Thus  $G$  is a spanning subgraph of  $G + e$ .

Many of the concepts that occur in graph theory and which we will investigate in detail later concern various ways in which one can “move about” in a graph. In particular, if we think of the vertices of a graph as locations and the edges as roads between certain pairs of locations, then the graph can be considered as modeling some community. There is a variety of kinds of trips that can be taken in the community.

Let's start at some vertex  $u$  of a graph  $G$ . If we proceed from  $u$  to a neighbor of  $u$  and then to a neighbor of that vertex, and so on, until we finally come to a stop at a vertex  $v$ , then we have just described a walk from  $u$  to  $v$  in  $G$ . More formally, a  $u - v$  **walk**  $W$  in  $G$  is a sequence of vertices in  $G$ , beginning with  $u$  and ending at  $v$  such that consecutive vertices in the sequence are adjacent, that is, we can express  $W$  as

$$W : u = v_0, v_1, \dots, v_k = v, \quad (1.1)$$

where  $k \geq 0$  and  $v_i$  and  $v_{i+1}$  are adjacent for  $i = 0, 1, 2, \dots, k-1$ . Each vertex  $v_i$  ( $0 \leq i \leq k$ ) and each edge  $v_i v_{i+1}$  ( $0 \leq i \leq k-1$ ) is said to lie on or belong to  $W$ . Notice that the definition of the walk  $W$  does not require the listed vertices to be distinct; in fact, even  $u$  and  $v$  are not required to be distinct. However, every two consecutive vertices in  $W$  are distinct since they are adjacent. If  $u = v$ , then the walk  $W$  is **closed**; while if  $u \neq v$ , then  $W$  is **open**. As we move from one vertex of  $W$  to the next, we are actually encountering or traversing edges of  $G$ , possibly traversing some edges of  $G$  more than once. The number of edges encountered in a walk (including multiple occurrences of an edge) is called the **length** of the walk. Thus the length of the walk  $W$  defined in (1.1) is  $k$ .

For the graph  $G$  of Figure 1.16,

$$W : x, y, w, y, v, w \quad (1.2)$$

is therefore a walk, indeed an  $x - w$  walk of length 5 (one less than the number of occurrences of vertices in the walk). A walk of length 0 is a **trivial walk**. So  $W : v$  is a trivial walk. (By this definition, those people who feel guilty about not exercising need not feel guilty any longer, as going for a daily “walk” just became easier.)

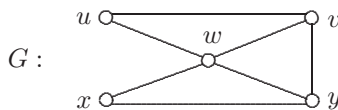


Figure 1.16: Illustrating walks in a graph

Provided we continue to proceed from a vertex to one of its neighbors (and eventually stop), there is essentially no conditions on a walk. However, there will be occasions when we want to place restrictions on certain types of walks.

Borrowing terminology from the Old West, we define a  $u - v$  **trail** in a graph  $G$  to be a  $u - v$  walk in which no edge is traversed more than once. Thus, the  $x - w$  walk  $W$  in (1.2) is *not* an  $x - w$  trail as the edge  $wy$  is repeated. On the other hand,

$$T : u, w, y, x, w, v \quad (1.3)$$

is a  $u - v$  trail in the graph  $G$  of Figure 1.16. Notice that this trail  $T$  repeats the vertex  $w$ . This is perfectly permissible. Although the definition of a trail stipulates that no edge can be repeated, no such condition is placed on vertices.

A  $u - v$  walk in a graph in which no vertices are repeated is a  $u - v$  **path**. While the  $u - v$  trail  $T$  in (1.3) is not a  $u - v$  path in the graph  $G$  of Figure 1.16 (since the vertex  $w$  is repeated),

$$P : u, w, y, v$$

is a  $u - v$  path. If no vertex in a walk is repeated (thereby producing a path), then no edge is repeated either. Hence every path is a trail.

If a  $u - v$  walk in a graph is followed by a  $v - w$  walk, then a  $u - w$  walk results. In particular, a  $u - v$  path followed by a  $v - w$  path is a  $u - w$  walk  $W$ , but not necessarily a  $u - w$  path, as vertices in  $W$  may be repeated. While not every walk is a path, if a graph contains a  $u - v$  walk, then it must also contain a  $u - v$  path. This is our first theorem. When a proof of a result is presented, the proof technique (or techniques used) is indicated. All of the proof techniques are reviewed in Appendix 3.

**Theorem 1.6** *If a graph  $G$  contains a  $u - v$  walk of length  $\ell$ , then  $G$  contains a  $u - v$  path of length at most  $\ell$ .*

**Proof.** [proof by contradiction] Among all  $u - v$  walks in  $G$ , let  $P : u = u_0, u_1, \dots, u_k = v$  be a  $u - v$  walk of smallest length  $k$ . Therefore,  $k \leq \ell$ . We claim that  $P$  is a  $u - v$  path. Assume, to the contrary, that this is not the case. Then some vertex of  $G$  must be repeated in  $P$ , say  $u_i = u_j$  for some  $i$  and  $j$  with  $0 \leq i < j \leq k$ . If we then delete the vertices  $u_{i+1}, u_{i+2}, \dots, u_j$  from  $P$ , we arrive at the  $u - v$  walk

$$u = u_0, u_1, \dots, u_{i-1}, u_i = u_j, u_{j+1}, \dots, u_k = v$$

whose length is less than  $k$ , which is impossible. Therefore, as claimed,  $P$  is a  $u - v$  path of length  $k \leq \ell$ . ■

A **circuit** in a graph  $G$  is a closed trail of length 3 or more. Hence a circuit begins and ends at the same vertex but repeats no edges. A circuit can be described by choosing any of its vertices as the beginning (and ending) vertex provided the vertices are listed in the same cyclic order. In a circuit, vertices can be repeated, in addition to the first and last. For example, in the graph  $G$  of Figure 1.16,

$$C : y, w, u, v, w, x, y \text{ or } C : x, y, w, u, v, w, x \text{ or } C : w, x, y, w, u, v, w \quad (1.4)$$

is a circuit. A circuit that repeats no vertex, except for the first and last, is a **cycle**. A  $k$ -**cycle** is a cycle of length  $k$ . A 3-cycle is also referred to as a **triangle**. A cycle of odd length is called an **odd cycle**; while, not surprisingly, a cycle of even length is called an **even cycle**. In the graph  $G$  of Figure 1.16, the circuit  $C$  in (1.4) is not a cycle, while

$$C' : x, y, v, w, x$$

is a cycle, namely a 4-cycle. If a vertex of a cycle is deleted, then a path is obtained. This is not necessarily true for circuits, however.

The vertices and edges of a trail, path, circuit, or cycle in a graph  $G$  form a subgraph of  $G$ , also called a **trail**, **path**, **circuit**, or **cycle**. Hence a path, for example, is used to describe both a manner of traversing certain vertices and edges of  $G$  and a subgraph consisting of those vertices and edges. The graph  $G$  of Figure 1.16 is shown again in Figure 1.17. Thus the subgraphs  $G_1, G_2, G_3, G_4$  of the graph  $G$  are a trail, path, circuit, and cycle, respectively.

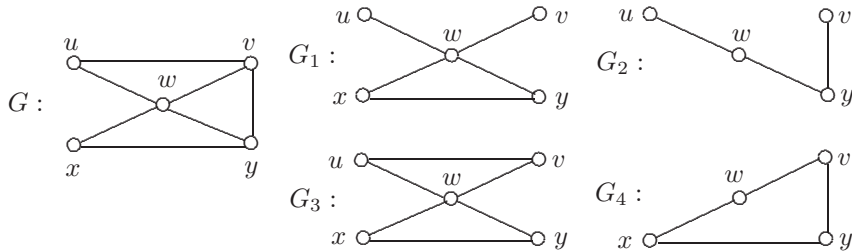


Figure 1.17: Trails, paths, circuits, and cycles as subgraphs of a graph

We will have a special interest in graphs  $G$  in which it is possible to travel from each vertex of  $G$  to any other vertex of  $G$ . If  $G$  contains a  $u - v$  path, then  $u$  and  $v$  are said to be **connected** and  $u$  is **connected to**  $v$  (and  $v$  is connected to  $u$ ). So, saying that  $u$  and  $v$  are connected only means that there is some  $u - v$  path in  $G$ ; it doesn't say that  $u$  and  $v$  are joined by an edge. Of course, if  $u$  is joined to  $v$ , then  $u$  is connected to  $v$  as well. A graph  $G$  is **connected** if every two vertices of  $G$  are connected, that is, if  $G$  contains a  $u - v$  path for every pair

$u, v$  of distinct vertices of  $G$ . By Theorem 1.6,  $G$  is connected if and only if  $G$  contains a  $u - v$  walk for every pair  $u, v$  of vertices of  $G$ . Since every vertex is connected to itself, the trivial graph is connected.

A graph  $G$  that is not connected is called **disconnected**. A connected subgraph of  $G$  that is not a proper subgraph of any other connected subgraph of  $G$  is a **component** of  $G$ . A graph  $G$  is then connected if and only if it has exactly one component. While the graph  $G$  of Figure 1.16 is connected, the graph  $H$  of Figure 1.18 is disconnected since, for example, there is no  $s - w$  path in  $H$ . There is no  $x - z$  path either. The graph  $H$  has three components, namely  $H_1$ ,  $H_2$ , and  $H_3$ .

For subgraphs  $G_1, G_2, \dots, G_k$ ,  $k \geq 2$ , of a graph  $G$ , with mutually disjoint vertex sets, we write  $G = G_1 \cup G_2 \cup \dots \cup G_k$  if every vertex and every edge of  $G$  belong to exactly one of these subgraphs. In particular, we write  $G = G_1 \cup G_2 \cup \dots \cup G_k$  if  $G_1, G_2, \dots, G_k$  are components of  $G$ . Therefore, we can write  $H = H_1 \cup H_2 \cup H_3$  for the graphs in Figure 1.18.

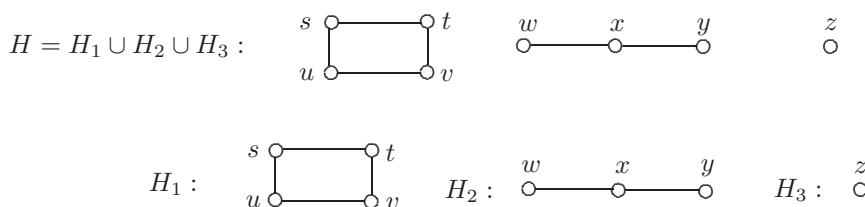


Figure 1.18: A disconnected graph and its components

Components can be looked by means of an equivalence relation. (Equivalence relations are reviewed in Appendix 2.1.)

**Theorem 1.7** *Let  $R$  be the relation defined on the vertex set of a graph  $G$  by  $u R v$ , where  $u, v \in V(G)$ , if  $u$  is connected to  $v$ , that is, if  $G$  contains a  $u - v$  path. Then  $R$  is an equivalence relation.*

**Proof.** [direct proof] It is immediate that  $R$  is reflexive and symmetric. It remains therefore only to show that  $R$  is transitive. Let  $u, v, w \in V(G)$  such that  $u R v$  and  $v R w$ . Hence  $G$  contains a  $u - v$  path  $P'$  and a  $v - w$  path  $P''$ . As we have seen earlier, following  $P'$  by  $P''$  produces a  $u - w$  walk  $W$ . By Theorem 1.6,  $G$  contains a  $u - w$  path and so  $u R w$ . ■

The equivalence relation described in Theorem 1.7 produces a partition of the vertex set of every graph  $G$  into equivalence classes. The subgraph of  $G$  induced by the vertices in an equivalence class is a component of  $G$ . Exercise 1.14 asks you to show this. As a consequence, we have the following:

*Each vertex and each edge of a graph  $G$  belong to exactly one component of  $G$ . This implies that if  $G$  is a disconnected graph and  $u$  and  $v$  are vertices belonging to different components of  $G$ , then  $uv \notin E(G)$ .*

The following theorem provides a sufficient condition for a graph of order at least 3 to be connected.

**Theorem 1.8** *Let  $G$  be a graph of order 3 or more. If  $G$  contains two distinct vertices  $u$  and  $v$  such that  $G - u$  and  $G - v$  are connected, then  $G$  itself is connected.*

**Proof.** [direct proof] Suppose that  $G$  contains distinct vertices  $u$  and  $v$  such that  $G - u$  and  $G - v$  are connected. To show that  $G$  itself is connected, we show that every two vertices of  $G$  are connected. Let  $x$  and  $y$  be two vertices of  $G$ . We consider two cases.

*Case 1.*  $\{x, y\} \neq \{u, v\}$ , say  $u \notin \{x, y\}$ . Then  $x$  and  $y$  are vertices in  $G - u$ . Since  $G - u$  is connected, there is an  $x - y$  path  $P$  in  $G - u$ . Hence  $P$  is in  $G$  and  $x$  and  $y$  are connected in  $G$ .

*Case 2.*  $\{x, y\} = \{u, v\}$ , say  $x = u$  and  $y = v$ . We show that  $u$  and  $v$  are connected in  $G$ . Since the order of  $G$  is at least 3, there is a vertex  $w$  in  $G$  such that  $w \neq u, v$ . Since  $G - v$  is connected,  $G - v$  contains a  $u - w$  path  $P'$ . Furthermore, since  $G - u$  is connected,  $G - u$  contains a  $w - v$  path  $P''$ . Therefore,  $P'$  followed by  $P''$  produces a  $u - v$  walk. By Theorem 1.6,  $G$  contains a  $u - v$  path and so  $u$  and  $v$  are connected in  $G$ . ■

If  $G$  is the disconnected graph consisting of two vertices  $u$  and  $v$  and no edges, then the subgraphs  $G - u$  and  $G - v$  are (trivially) connected. Therefore, in Theorem 1.8, it is essential that the order of the graph under consideration be at least 3.

If  $u$  and  $v$  are vertices in a connected graph  $G$ , then there must be a  $u - v$  path in  $G$ . However, it is quite possible that  $G$  contains several  $u - v$  paths. For example, in the graph  $G$  of Figure 1.16, all of the following are  $u - y$  paths:

$$P' : u, v, y \quad P'' : u, w, v, y \quad P''' : u, v, w, x, y.$$

The length of  $P'$  is 2, the length of  $P''$  is 3, and the length of  $P'''$  is 4. There is no  $u - y$  path of length 1 in this graph since  $u$  and  $y$  are not adjacent, and there are no  $u - y$  paths of length 5 or more as  $G$  only has five vertices.

Let  $G$  be a connected graph of order  $n$ , and let  $u$  and  $v$  be two vertices of  $G$ . The **distance** between  $u$  and  $v$  is the smallest length of any  $u - v$  path in  $G$  and is denoted by  $d_G(u, v)$  or simply  $d(u, v)$  if the graph  $G$  under consideration is clear. Hence if  $d(u, v) = k$ , then there exists a  $u - v$  path

$$P : u = v_0, v_1, \dots, v_k = v \tag{1.5}$$

of length  $k$  in  $G$ , but no  $u - v$  path of smaller length exists in  $G$ . A  $u - v$  path of length  $d(u, v)$  is called a  $u - v$  **geodesic**. In fact, since the path  $P$  in (1.5) is a  $u - v$  geodesic, not only is  $d(u, v) = d(u, v_k) = k$ , but  $d(u, v_i) = i$  for every  $i$  with  $0 \leq i \leq k$ . Exercise 1.16 asks you to verify this. If  $u = v$ , then  $d(u, v) = 0$ . If  $uv \in E(G)$ , then  $d(u, v) = 1$ . In general,  $0 \leq d(u, v) \leq n - 1$

for every two vertices  $u$  and  $v$  (distinct or not) in a connected graph of order  $n$ . For the vertices  $u$  and  $y$  in the graph  $G$  of Figure 1.16,  $d(u, y) = 2$ . If  $G$  is disconnected, then there are some pairs  $x, y$  of distinct vertices of  $G$  such that there is no  $x - y$  path in  $G$ . In this case,  $d(x, y)$  is not defined.

At times, it is useful to visualize the vertices of a connected graph according to their distances from a given vertex. The graph  $H$  of Figure 1.19(a) is redrawn in Figure 1.19(b) to indicate those vertices at a given distance from the vertex  $t$ . The vertex  $t$  (the only vertex whose distance from  $t$  is 0) is drawn at the top. The vertices one level down are the neighbors of  $t$ . The next level consists of those vertices whose distance from  $t$  is 2, and so on. Observe that two adjacent vertices must either belong to the same level or to neighboring levels.

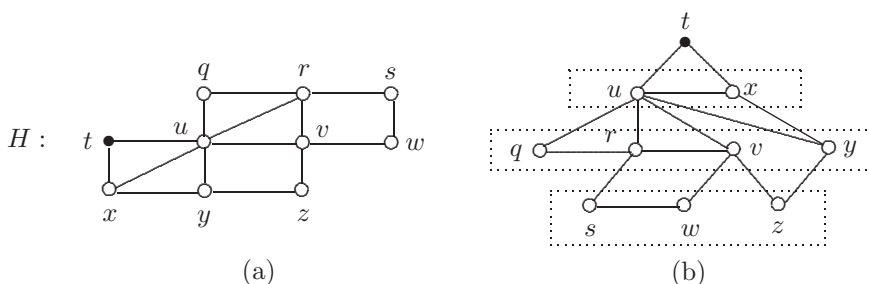


Figure 1.19: Distances from a given vertex

The greatest distance between any two vertices of a connected graph  $G$  is called the **diameter** of  $G$  and is denoted by  $\text{diam}(G)$ . The diameter of the graph  $H$  of Figure 1.19 is 3. The path  $P' : y, u, r, s$  is a  $y - s$  geodesic whose length is  $\text{diam}(H)$ .

If  $G$  is a connected graph such that  $d(u, v) = \text{diam}(G)$  and  $w \neq u, v$ , then no  $u - w$  geodesic can contain  $v$ , for otherwise  $d(u, w) > d(u, v) = \text{diam}(G)$ , which is impossible.

Let's return to Theorem 1.8, where we proved that if a graph  $G$  of order 3 contains two distinct vertices  $u$  and  $v$  such that  $G - u$  and  $G - v$  are connected, then  $G$  is connected. Actually, the converse of this theorem is also true; that is, if  $G$  is a connected graph of order at least 3, then  $G$  must contain two vertices  $u$  and  $v$  such that  $G - u$  and  $G - v$  are both connected. We are now in a position to prove this theorem as well.

**Theorem 1.9** *If  $G$  is a connected graph of order 3 or more, then  $G$  contains two distinct vertices  $u$  and  $v$  such that  $G - u$  and  $G - v$  are connected.*

**Proof.** [proof by contradiction] Let  $u$  and  $v$  be two vertices of  $G$  such that  $d(u, v) = \text{diam}(G)$ . We claim that  $G - u$  and  $G - v$  are both connected. Suppose that this is not the case. Then at least one of  $G - u$  and  $G - v$  is disconnected, say  $G - v$  is disconnected. Therefore,  $G - v$  contains two vertices  $x$  and  $y$  that



are not connected in  $G - v$ . However, since  $G$  is connected, the vertices  $u$  and  $x$  are connected in  $G$ , as are  $u$  and  $y$ .

Let  $P'$  be an  $x - u$  geodesic in  $G$  and let  $P''$  be a  $u - y$  geodesic in  $G$ . Since  $d_G(u, v) = \text{diam}(G)$ , the vertex  $v$  cannot lie on either  $P'$  or on  $P''$ , so  $P'$  and  $P''$  are paths in  $G - v$ . The path  $P'$  followed by  $P''$  produces an  $x - y$  walk  $W$  in  $G - v$ . By Theorem 1.6,  $G - v$  contains an  $x - y$  path and so  $x$  and  $y$  are connected in  $G - v$ . This is a contradiction. ■

Theorem 1.9 gives a property that every connected graph of order at least 3 must have. That is, Theorem 1.9 provides a *necessary condition* for a graph to be connected. Actually, Theorem 1.9 is true even if the order of  $G$  is 2, but we stated Theorem 1.9 as we did so we could combine Theorems 1.8 and 1.9 into a single *necessary and sufficient condition* for a graph to be connected, which we state next.

**Theorem 1.10** *Let  $G$  be a graph of order 3 or more. Then  $G$  is connected if and only if  $G$  contains two distinct vertices  $u$  and  $v$  such that  $G - u$  and  $G - v$  are connected.*

### Exercises for Section 1.2

- 1.11 Let  $G$  be the graph of Figure 1.20, let  $X = \{e, f\}$ , where  $e = ru$  and  $f = vw$ , and let  $U = \{u, w\}$ . Draw the subgraphs  $G - X$  and  $G - U$  of  $G$ .

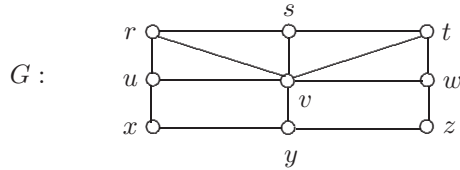


Figure 1.20: The graph  $G$  in Exercises 1.11 and 1.12

- 1.12 For the graph  $G$  of Figure 1.20, give an example of each of the following or explain why no such example exists.
- An  $x - y$  walk of length 6.
  - A  $v - w$  trail that is not a  $v - w$  path.
  - An  $r - z$  path of length 2.
  - An  $x - z$  path of length 3.
  - An  $x - t$  path of length  $d(x, t)$ .
  - A circuit of length 10.
  - A cycle of length 8.
  - A geodesic whose length is  $\text{diam}(G)$ .

- 1.13 Draw a graph  $F = F_1 \cup F_2 \cup F_3 \cup F_4$ , where  $F_i$  is a connected graph of order  $i$  for  $1 \leq i \leq 4$ .
- What is the order and size of your graph  $F$ ?
  - What is the smallest possible size and largest possible size for such a graph  $F$ ?
- 1.14 For a graph  $G$ , a component of  $G$  has been defined as (1) a connected subgraph of  $G$  that is not a proper subgraph of any other connected subgraph of  $G$  and has been described as (2) a subgraph of  $G$  induced by the vertices in an equivalence class resulting from the equivalence relation defined in Theorem 1.7. Show that these two interpretations of components are equivalent.
- 1.15 Draw all connected graphs of order 5 in which the distance between every two distinct vertices is odd. Explain why you know that you have drawn all such graphs.
- 1.16 Let  $P : u = v_0, v_1, \dots, v_k = v$ ,  $k \geq 1$ , be a  $u - v$  geodesic in a connected graph  $G$ . Prove that  $d(u, v_i) = i$  for each integer  $i$  with  $1 \leq i \leq k$ .
- 1.17 (a) Prove that if  $P$  and  $Q$  are two longest paths in a connected graph, then  $P$  and  $Q$  have at least one vertex in common.
- (b) Prove or disprove: Let  $G$  be connected graph of diameter  $k$ . If  $P$  and  $Q$  are two geodesics of length  $k$  in  $G$ , then  $P$  and  $Q$  have at least one vertex in common.
- 1.18 A graph  $G$  of order 12 has vertex set  $V(G) = \{c_1, c_2, \dots, c_{12}\}$  for the twelve configurations in Figure 1.4. A “move” on this checkerboard corresponds to moving a single coin to an unoccupied square, where
- the gold coin can only be moved horizontally or diagonally,
  - the silver coin can only be moved vertically or diagonally.

Two vertices  $c_i$  and  $c_j$  ( $i \neq j$ ) are adjacent if it is possible to move  $c_i$  to  $c_j$  by a single move.

- What vertices are adjacent to  $c_1$  in  $G$ ?
  - What vertices are adjacent to  $c_2$  in  $G$ ?
  - Draw the subgraph of  $G$  induced by  $\{c_2, c_6, c_9, c_{11}\}$ .
  - Give an example of a  $c_1 - c_7$  path in  $G$ .
- 1.19 Theorem 1.10 states that a graph  $G$  of order 3 or more is connected if and only if  $G$  contains two distinct vertices  $u$  and  $v$  such that  $G - u$  and  $G - v$  are connected. Based on this, one might suspect that the following statement is true. *Every connected graph  $G$  of order 4 or more contains three distinct vertices  $u, v$ , and  $w$  such that  $G - u$ ,  $G - v$ , and  $G - w$  are connected.* Is it?

- 1.20 (a) Let  $u$  and  $v$  be distinct vertices in a connected graph  $G$ . There may be several connected subgraphs of  $G$  containing  $u$  and  $v$ . What is the minimum size of a connected subgraph of  $G$  containing  $u$  and  $v$ ? Explain your answer.
- (b) Does the question in (a) suggest another question to you?

## 1.3 Common Classes of Graphs

As we continue to study graphs, we will see that there are certain graphs that are encountered often and it is useful to be familiar with them. In many instances, there is special notation reserved for these graphs.

We have already seen that paths and cycles are certain kinds of walks in graphs. These terms are also used to describe certain kinds of graphs. If the vertices of a graph  $G$  of order  $n$  can be labeled (or relabeled)  $v_1, v_2, \dots, v_n$  so that its edges are  $v_1v_2, v_2v_3, \dots, v_{n-1}v_n$ , then  $G$  is called a **path**; while if the vertices of a graph  $G$  of order  $n \geq 3$  can be labeled (or relabeled)  $v_1, v_2, \dots, v_n$  so that its edges are  $v_1v_2, v_2v_3, \dots, v_{n-1}v_n$ , and  $v_1v_n$ , then  $G$  is called a **cycle**. A graph that is a path of order  $n$  is denoted by  $P_n$ , while a graph that is a cycle of order  $n \geq 3$  is denoted by  $C_n$ . Several paths and cycles are shown in Figure 1.21.

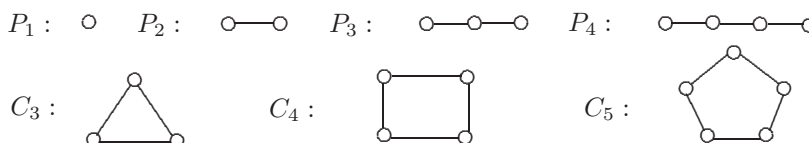


Figure 1.21: Paths and cycles

A graph  $G$  is **complete** if every two distinct vertices of  $G$  are adjacent. A complete graph of order  $n$  is denoted by  $K_n$ . Therefore,  $K_n$  has the maximum possible size for a graph with  $n$  vertices. Since every two distinct vertices of  $K_n$  are joined by an edge, the number of pairs of vertices in  $K_n$  is  $\binom{n}{2}$  and so

$$\text{the size of } K_n \text{ is } \binom{n}{2} = \frac{n(n-1)}{2}. \quad (1.6)$$

Therefore, the complete graph  $K_3$  has three edges,  $K_4$  has six edges, and  $K_5$  has ten edges. The first five complete graphs are shown in Figure 1.22. Notice that  $P_1$  and  $K_1$  represent the same graph, as do  $P_2$  and  $K_2$ , as well as  $C_3$  and  $K_3$ . Although there are edges that cross in the drawings of  $K_4$  and  $K_5$ , the points of intersection do not represent vertices.

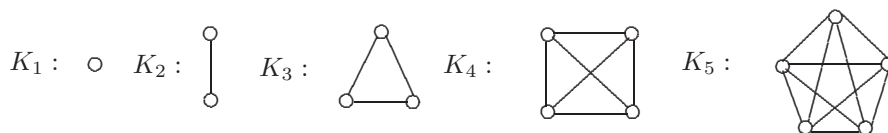
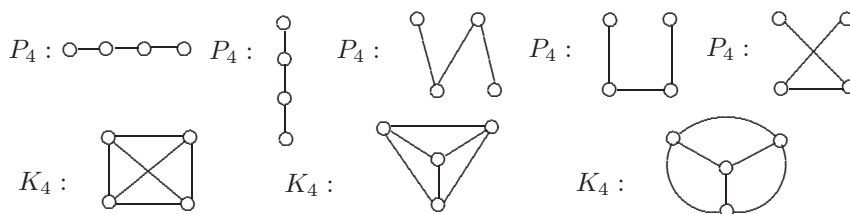
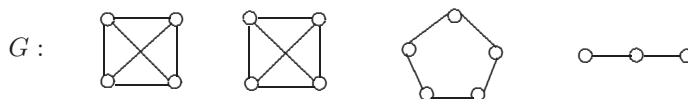


Figure 1.22: Complete graphs

The graphs that are drawn in Figures 1.21 and 1.22 bring up some points that need to be discussed. Although we have attempted to draw these graphs in a manner that makes them easy to visualize, this is certainly not a requirement when drawing a graph, as its vertices can be placed in any convenient location. Figure 1.23 shows a variety of ways to draw the path  $P_4$  and the complete graph  $K_4$ .

Figure 1.23: The graphs  $P_4$  and  $K_4$ 

Since the disconnected graph  $G$  in Figure 1.24 has two components that are complete graphs of order 4, one that is  $C_5$ , and one that is  $P_3$ , we write this graph as  $G = 2K_4 \cup C_5 \cup P_3$ .

Figure 1.24: The graph  $G = 2K_4 \cup C_5 \cup P_3$ 

The **complement**  $\overline{G}$  of a graph  $G$  is that graph whose vertex set is  $V(G)$  and such that for each pair  $u, v$  of vertices of  $G$ ,  $uv$  is an edge of  $\overline{G}$  if and only if  $uv$  is not an edge of  $G$ . Observe that if  $G$  is a graph of order  $n$  and size  $m$ , then  $\overline{G}$  is a graph of order  $n$  and size  $\binom{n}{2} - m$ . The graph  $\overline{K}_n$  then has  $n$  vertices and no edges; it is called the **empty graph** of order  $n$ . Therefore, empty graphs have empty edge sets. In fact, if  $G$  is any graph of order  $n$ , then  $G - E(G)$  is the empty graph  $\overline{K}_n$ . By definition, no graph can have an empty vertex set. A graph  $H$  and its complement are shown in Figure 1.25. Both of these graphs are connected. Although a graph and its complement need not both be connected, at least one must be connected.

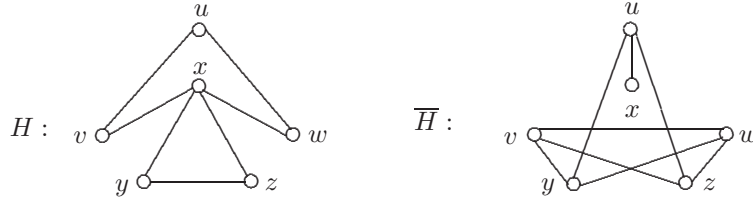


Figure 1.25: A graph and its complement

**Theorem 1.11** *If  $G$  is a disconnected graph, then  $\overline{G}$  is connected.*

**Proof.** [direct proof] Since  $G$  is disconnected,  $G$  contains two or more components. Let  $u$  and  $v$  be two vertices of  $\overline{G}$ . We show that  $u$  and  $v$  are connected in  $\overline{G}$ . If  $u$  and  $v$  belong to different components of  $G$ , then  $u$  and  $v$  are not adjacent in  $G$  and so  $u$  and  $v$  are adjacent in  $\overline{G}$ . Hence  $\overline{G}$  contains a  $u-v$  path of length 1. Suppose next that  $u$  and  $v$  belong to the same component of  $G$ . Let  $w$  be a vertex of  $G$  that belongs to a different component of  $G$ . Then  $uw, vw \notin E(G)$ , implying that  $uw, vw \in E(\overline{G})$  and so  $u, w, v$  is a  $u-v$  path in  $\overline{G}$ . ■

We now turn to graphs whose vertex sets can be partitioned in special ways. A graph  $G$  is a **bipartite graph** if  $V(G)$  can be partitioned into two subsets  $U$  and  $W$ , called **partite sets**, such that every edge of  $G$  joins a vertex of  $U$  and a vertex of  $W$ . It's not always easy to tell at a glance whether a graph is bipartite. For example, the connected graphs  $G_1$  and  $G_2$  of Figure 1.26 are bipartite, as every edge of  $G_1$  joins a vertex of  $U_1 = \{u_1, x_1, y_1\}$  and a vertex of  $W_1 = \{v_1, w_1\}$ , while every edge of  $G_2$  joins a vertex of  $U_2 = \{u_2, w_2, y_2\}$  and a vertex of  $W_2 = \{v_2, x_2, z_2\}$ . The bipartite nature of these graphs is illustrated in Figure 1.26. By letting  $U = U_1 \cup U_2$  and  $W = W_1 \cup W_2$ , we see that every edge of  $G = G_1 \cup G_2$  joins a vertex of  $U$  and a vertex of  $W$ . This illustrates the observation that a graph is bipartite if and only if each of its components is bipartite.

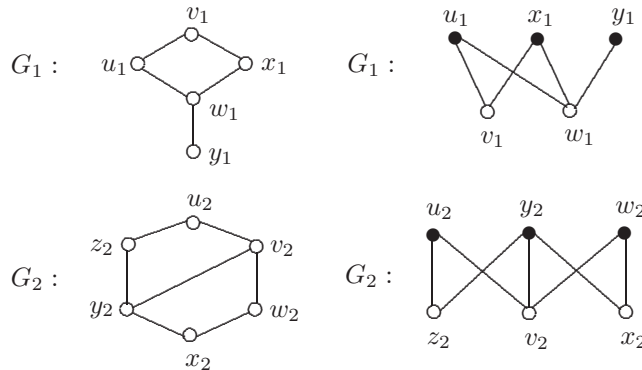


Figure 1.26: Bipartite graphs

Certainly not every graph is bipartite. For example, consider the 5-cycle  $C_5$  in Figure 1.27. If  $C_5$  were bipartite, then its vertex set can be partitioned into two sets  $U$  and  $W$  such that every edge of  $C_5$  joins a vertex of  $U$  and a vertex of  $W$ . The vertex  $v_1$  must belong to either  $U$  or  $W$ , say  $v_1 \in U$ . Since  $v_1v_2$  is an edge of  $C_5$ , it follows that  $v_2 \in W$ . Since  $v_2v_3$  is an edge of  $C_5$ , it follows that  $v_3 \in U$ . Similarly,  $v_4 \in W$  and  $v_5 \in U$ . However,  $v_1, v_5 \in U$  and  $v_1v_5$  is an edge of  $C_5$ . This is a contradiction. Therefore,  $C_5$  is not bipartite. In fact, no odd cycle is bipartite. Indeed, any graph that contains an odd cycle is not bipartite. The converse is true as well, which may come as a surprise.

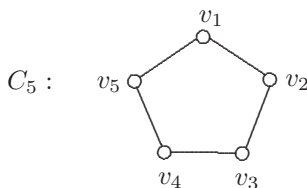


Figure 1.27: A 5-cycle: A graph that is not bipartite

**Theorem 1.12** *A nontrivial graph  $G$  is a bipartite graph if and only if  $G$  contains no odd cycles.*

**Proof.** [proof by contrapositive, proof by contradiction] We have already seen that if a graph contains an odd cycle, then it's not bipartite. To prove the converse, let  $G$  be a nontrivial graph having no odd cycles. We show that  $G$  is bipartite. Because of our earlier observation that a graph is bipartite if and only if each of its components is bipartite, we may assume that  $G$  is connected. Let  $u$  be any vertex of  $G$ , let  $U$  consist of all vertices of  $G$  whose distance from  $u$  is even, and let  $W$  consist of all vertices whose distance from  $u$  is odd. Thus  $\{U, W\}$  is a partition of  $V(G)$ . Since  $d(u, u) = 0$ , it follows that  $u \in U$ . We claim that every edge of  $G$  joins a vertex of  $U$  and a vertex of  $W$ .

Assume, to the contrary, that there exist two adjacent vertices in  $U$  or two adjacent vertices in  $W$ . Since these two situations are similar, we will assume that there are vertices  $v$  and  $w$  in  $W$  such that  $vw \in E(G)$ . Since  $d(u, v)$  and  $d(u, w)$  are both odd,  $d(u, v) = 2s + 1$  and  $d(u, w) = 2t + 1$  for nonnegative integers  $s$  and  $t$ . Let  $P' : u = v_0, v_1, \dots, v_{2s+1} = v$  be a  $u - v$  geodesic and let  $P'' : u = w_0, w_1, \dots, w_{2t+1} = w$  be a  $u - w$  geodesic in  $G$ . Certainly,  $P'$  and  $P''$  have their initial vertex  $u$  in common, but they may have other vertices in common as well. Among the vertices  $P'$  and  $P''$  have in common, let  $x$  be the last vertex. Perhaps  $x = u$ . In any case,  $x = v_i$  for some integer  $i \geq 0$ . Thus  $d(u, v_i) = i$ . Since  $x$  is on  $P''$  and  $w_i$  is the only vertex of  $P''$  whose distance from  $u$  is  $i$ , it follows that  $x = w_i$ . So  $x = v_i = w_i$ . However then,  $C : v_i, v_{i+1}, \dots, v_{2s+1}, w_{2t+1}, w_{2t}, \dots, w_i = v_i$  is a cycle of length

$$[(2s + 1) - i] + [(2t + 1) - i] + 1 = 2s + 2t - 2i + 3 = 2(s + t - i + 1) + 1$$

and so  $C$  is an odd cycle, which is a contradiction. ■

We know that if  $G$  is a bipartite graph, then  $V(G)$  can be partitioned into two subsets  $U$  and  $W$ , called partite sets, such that every edge of  $G$  joins a vertex of  $U$  and a vertex of  $W$ . However, this does not mean that every vertex of  $U$  is adjacent to every vertex of  $W$ . If this does happen, however, then we call  $G$  a **complete bipartite graph**. A complete bipartite graph with  $|U| = s$  and  $|W| = t$  is denoted by  $K_{s,t}$  or  $K_{t,s}$ . If either  $s = 1$  or  $t = 1$ , then  $K_{s,t}$  is a **star**. Several complete bipartite graphs are shown in Figure 1.28, including the star  $K_{1,3}$ . Observe that  $K_{2,2}$  is the same graph as  $C_4$ , although it is certainly not drawn the same way that we drew  $C_4$  in Figure 1.21. When two graphs  $G$  and  $H$  are the same except possibly for the way that they're drawn or their vertices are labeled, then we write  $G \cong H$ . (The technical term for this is that these graphs are isomorphic. We'll discuss this in Chapter 3.) If the structures of  $G$  and  $H$  are different, then we write  $G \not\cong H$ .

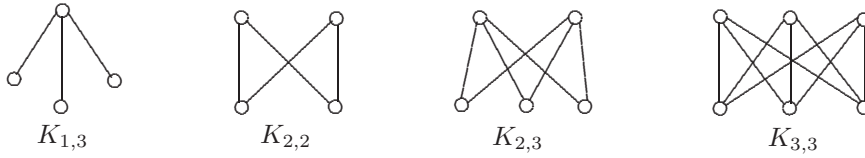


Figure 1.28: Complete bipartite graphs

Bipartite graphs belong to a more general class of graphs. A graph  $G$  is a  **$k$ -partite graph** if  $V(G)$  can be partitioned into  $k$  subsets  $V_1, V_2, \dots, V_k$  (once again called **partite sets**) such that  $uv$  is an edge of  $G$  if  $u$  and  $v$  belong to different partite sets. If, in addition, every two vertices in different partite sets are joined by an edge, then  $G$  is a **complete  $k$ -partite graph**. If  $|V_i| = n_i$  for  $1 \leq i \leq k$ , then we denote this complete  $k$ -partite graph by  $K_{n_1, n_2, \dots, n_k}$ . The complete  $k$ -partite graphs are also referred to as **complete multipartite graphs**. If  $n_i = 1$  for every  $i$  ( $1 \leq i \leq k$ ), then  $K_{n_1, n_2, \dots, n_k}$  is the complete graph  $K_k$ . Complete 2-partite graphs are thus complete bipartite graphs. Several complete multipartite graphs are shown in Figure 1.29.

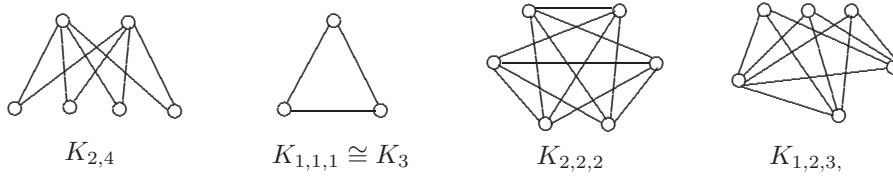


Figure 1.29: Complete multipartite graphs

There are several ways to produce a new graph from a given pair of graphs. For two vertex-disjoint graphs  $G$  and  $H$ , we have already mentioned that  $G \cup H$  is that (disconnected) graph with vertex set  $V(G) \cup V(H)$  and edge set

$E(G) \cup E(H)$ . The **join**  $G + H$  consists of  $G \cup H$  and all edges joining a vertex of  $G$  and a vertex of  $H$ . The join of  $P_3$  and  $K_2$  is shown in Figure 1.30.

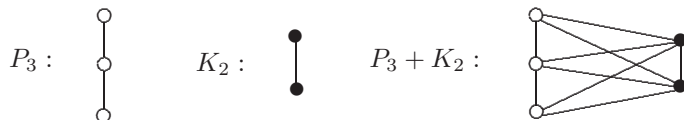


Figure 1.30: The join of two graphs

For two (not necessarily vertex-disjoint) graphs  $G$  and  $H$ , the **Cartesian product**  $G \times H$  has vertex set  $V(G \times H) = V(G) \times V(H)$ , that is, every vertex of  $G \times H$  is an ordered pair  $(u, v)$ , where  $u \in V(G)$  and  $v \in V(H)$ . Two distinct vertices  $(u, v)$  and  $(x, y)$  are adjacent in  $G \times H$  if either (1)  $u = x$  and  $vy \in E(H)$  or (2)  $v = y$  and  $ux \in E(G)$ . Figure 1.31 shows the Cartesian product of  $P_3$  and  $K_2$ .

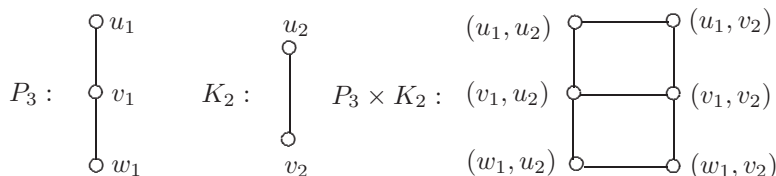


Figure 1.31: The Cartesian product of two graphs

Some additional comments about Cartesian products of graphs are useful. First, the definition of Cartesian product tells us that the order in which the graphs  $G$  and  $H$  are written is structurally irrelevant, that is,  $G \times H$  and  $H \times G$  are the same graph, that is, they are isomorphic graphs.

There is an informal way of drawing the graph  $G \times H$  (or  $H \times G$ ) that doesn't require us to label the vertices. Replace each vertex  $x$  of  $G$  by a copy  $H_x$  of the graph  $H$ . Let  $u$  and  $v$  be two vertices of  $G$ . If  $u$  and  $v$  are adjacent in  $G$ , then we join corresponding vertices of  $H_u$  and  $H_v$  by an edge. If  $u$  and  $v$  are not adjacent in  $G$ , then we add no edges between  $H_u$  and  $H_v$ . This is illustrated in Figure 1.32.

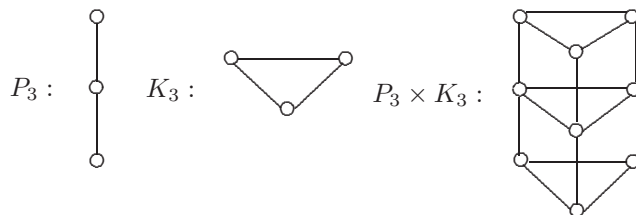


Figure 1.32: The Cartesian product of two graphs



Notice that  $K_2 \times K_2$  is the 4-cycle. The graph  $C_4 \times K_2$  is often denoted by  $Q_3$  and is called the **3-cube**. More generally, we define  $Q_1$  to be  $K_2$  and for  $n \geq 2$ , define  $Q_n$  to be  $Q_{n-1} \times K_2$ . The graphs  $Q_n$  are then called  **$n$ -cubes** or **hypercubes**. The  $n$ -cube can also be defined as that graph whose vertex set is the set of ordered  $n$ -tuples of 0s and 1s (commonly called  **$n$ -bit strings**) and where two vertices are adjacent if their ordered  $n$ -tuples differ in exactly one position (coordinate). The  $n$ -cubes for  $n = 1, 2, 3$  are shown in Figure 1.33, where their vertices are labeled by  $n$ -bit strings.

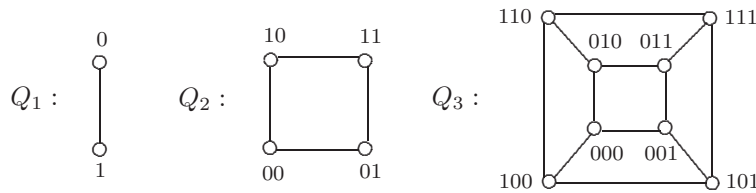


Figure 1.33: The  $n$ -cubes for  $1 \leq n \leq 3$

### Exercises for Section 1.3

- 1.21 Draw the graph  $3P_4 \cup 2C_4 \cup K_4$ .
- 1.22 Let  $G$  be a disconnected graph. By Theorem 1.11,  $\overline{G}$  is connected. Prove that if  $u$  and  $v$  are any two vertices of  $\overline{G}$ , then  $d_{\overline{G}}(u, v) = 1$  or  $d_{\overline{G}}(u, v) = 2$ . Therefore, if  $G$  is a disconnected graph, then  $\text{diam}(\overline{G}) \leq 2$ .
- 1.23 Consider the following question: For a given positive integer  $k$ , does there exist a connected graph  $G$  whose complement  $\overline{G}$  is also connected and contains four distinct vertices  $u, v, x, y$  for which  $d_G(u, v) = k = d_{\overline{G}}(x, y)$ ?
  - (a) Show that the answer to this question is yes if  $k = 1$  or  $k = 2$ .
  - (b) Find the largest value of  $k$  for which the answer to this question is yes.
- 1.24 Determine whether the graphs  $G_1$  and  $G_2$  of Figure 1.34 are bipartite. If a graph is bipartite, then redraw it indicating the partite sets; if not, then give an explanation as to why the graph is not bipartite.
- 1.25 Let  $G$  be a graph of order 5 or more. Prove that at most one of  $G$  and  $\overline{G}$  is bipartite.
- 1.26 Suppose that the vertex set of a graph  $G$  is a (finite) set of integers. Two vertices  $x$  and  $y$  are adjacent if  $x + y$  is odd. To which well-known class of graphs is  $G$  a member? Explain.

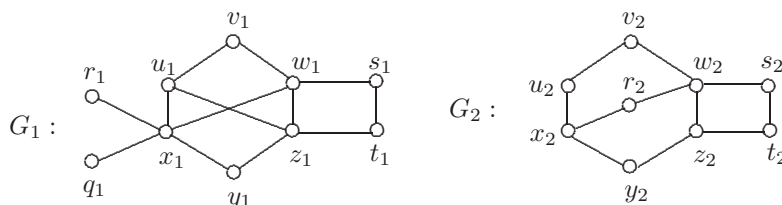


Figure 1.34: Graphs in Exercise 1.24

- 1.27 For the following pairs  $G, H$  of graphs, draw  $G + H$  and  $G \times H$ .
- (a)  $G = K_5$  and  $H = K_2$ .
  - (b)  $G = \overline{K}_5$  and  $H = \overline{K}_3$ .
  - (c)  $G = C_5$  and  $H = K_1$ .
- 1.28 We have seen that for  $n \geq 1$ , the  $n$ -cube  $Q_n$  is that graph whose vertex set is the set of  $n$ -bit strings, where two vertices of  $S_n$  are adjacent if they differ in exactly one coordinate.
- (a) For  $n \geq 2$ , define the graph  $R_n$  to be that graph whose vertex set is the set of  $n$ -bit strings, where two vertices of  $R_n$  are adjacent if they differ in exactly two coordinates. Draw  $R_2$  and  $R_3$ .
  - (b) For  $n \geq 3$ , define the graph  $S_n$  to be that graph whose vertex set is the set of  $n$ -bit strings, where two vertices of  $R_n$  are adjacent if they differ in exactly three coordinates. Draw  $S_2$  and  $S_3$ .

## 1.4 Multigraphs and Digraphs

There are occasions when graphs may not be an appropriate model for a problem we are investigating. We now describe two variations of graphs that we will encounter from time to time. In a graph, two vertices are either adjacent or they are not, that is, two vertices are joined by one edge or no edges. A **multigraph**  $M$  consists of a finite nonempty set  $V$  of vertices and a set  $E$  of edges, where every two vertices of  $M$  are joined by a finite number of edges (possibly zero). If two or more edges join the same pair of (distinct) vertices, then these edges are called **parallel edges**. In a multigraph, an edge is also permitted to join a vertex to itself. Such an edge is called a **loop**. If a loop  $e$  joins a vertex  $v$  to itself, then  $e$  is said to be a loop at  $v$ . There can be any finite number of loops at the same vertex in a multigraph. Four multigraphs are shown in Figure 1.35.

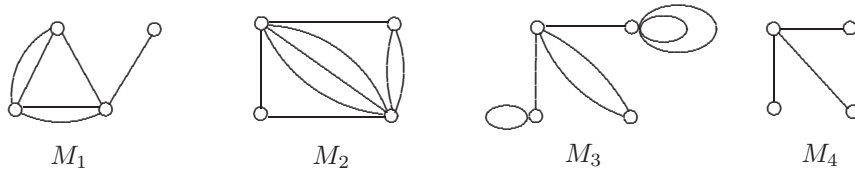


Figure 1.35: Multigraphs

The multigraph  $M_4$  illustrates the observation that every graph is a multigraph, although the converse is certainly not true.

If  $M$  is a multigraph with vertex set  $V$ , then it is no longer appropriate to regard an edge of  $M$  as a 2-element subset of  $V$  as we must somehow indicate the multiplicity of the edge and make allowances for the existence of loops.

Let's return to Example 1.2 where we considered the set  $S = \{2, 3, 5, 8, 13, 21\}$  as well as those pairs of integers of  $S$  whose sum or difference (in absolute value) belongs to  $S$ . The graph  $H$  of Figure 1.2 models this situation. In  $H$  there is an edge joining the vertices 3 and 5, indicating that  $3 + 5 \in S$  or  $|3 - 5| \in S$ . In this case, however, *both*  $3 + 5 \in S$  and  $|3 - 5| \in S$ , but there is no way of knowing this from  $H$ . The multigraph  $M$  of Figure 1.36 supplies this information, however. Even in this case, the existence of a single edge between a pair  $i, j$  of vertices doesn't tell us whether  $i + j \in S$  or  $|i - j| \in S$ ; it only tells us that one of these occurs. Thus the multigraph  $M$  of Figure 1.36 is a better model of this situation.

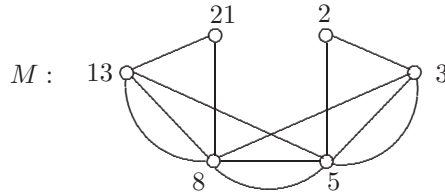


Figure 1.36: A multigraph

A **digraph** (or **directed graph**)  $D$  is a finite nonempty set  $V$  of objects called **vertices** together with a set  $E$  of *ordered pairs* of distinct vertices. The elements of  $E$  are called **directed edges** or **arcs**. If  $(u, v)$  is a directed edge, then we indicate this in a diagram representing  $D$  by drawing a directed line segment or curve from  $u$  to  $v$ . Then  $u$  is said to be **adjacent to**  $v$ , and  $v$  is **adjacent from**  $u$ . The vertices  $u$  and  $v$  are also said to be **incident with** the directed edge  $(u, v)$ . Arcs  $(u, v)$  and  $(v, u)$  may both be present in some directed graph. If, in the definition of digraph, for each pair  $u, v$  of distinct vertices, at most one of  $(u, v)$  and  $(v, u)$  is a directed edge, then the resulting digraph is an **oriented graph**. Thus an oriented graph  $D$  is obtained by assigning a direction to each edge of some *graph*  $G$ . The digraph  $D$  is also called an **orientation** of  $G$ . Figure 1.37 shows two digraphs  $D_1$  and  $D_2$ , where  $D_2$  is an oriented graph but  $D_1$  is not.

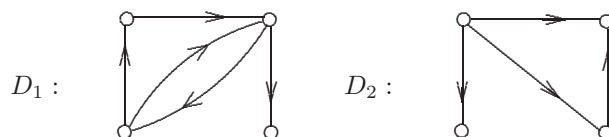


Figure 1.37: Digraphs

Next, we return to Example 1.3, where we considered twelve configurations of two coins (one silver, one gold), which were denoted by  $c_1, c_2, \dots, c_{12}$ . Now, we say that  $c_i$  can be transformed into  $c_j$  if  $c_j$  can be obtained by moving one of the coins in  $c_i$  to the right or up. Modeling this situation requires a digraph, namely, the digraph  $D$  shown in Figure 1.38, which is an oriented graph.

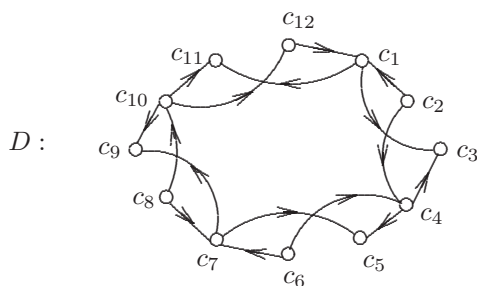
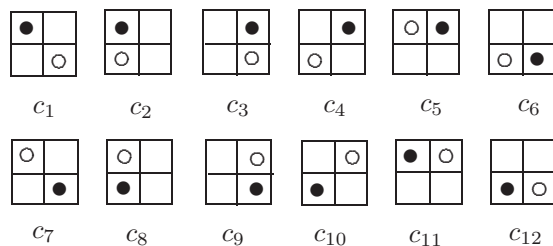


Figure 1.38: Modeling twelve configurations by a digraph

#### Exercises for Section 1.4

- 1.29 (a) Let  $S = \{2, 3, 4, 7, 11, 13\}$ . Construct the multigraph  $M$  whose vertex set is  $S$  and where  $ij$  is an edge for distinct elements  $i$  and  $j$  in  $S$  whenever  $i + j \in S$ , and  $ij$  is an edge whenever  $|i - j| \in S$ . In other words,  $i$  and  $j$  are joined by two edges if both  $i + j \in S$  and  $|i - j| \in S$ .
- (b) How is the answer in (a) affected if we remove the word “distinct”?
- 1.30 Consider the twelve configurations  $c_i$ ,  $1 \leq i \leq 12$ , in Figure 1.38. Draw the digraph  $D$ , where  $V(D) = \{c_1, c_2, \dots, c_{12}\}$  and where  $(c_i, c_j)$  is a directed

edge of  $D$  if it is possible to obtain  $c_j$  by rotating the configuration  $c_i$  either  $90^\circ$  or  $180^\circ$  clockwise about the midpoint of the checkerboard.

- 1.31 Using the twelve configurations in Figure 1.38, define a transformation different from the one described in Exercise 1.30 which can be modeled by a digraph but not by a graph.
- 1.32 Let  $S$  and  $A$  be two finite nonempty sets of integers. Define a digraph  $D$  with  $V(D) = A$ , where  $(x, y)$  is an arc of  $D$  if  $x \neq y$  and  $y - x \in S$ .
- (a) Draw the digraph  $D$  for  $A = \{0, 1, 2, 3, 4\}$  and  $S = \{-2, 1, 2, 4\}$ .
  - (b) What can be said about  $D$  if  $A$  and  $S$  consist only of odd integers?
  - (c) How can the question in (b) be generalized?
  - (d) If  $|A| = |S| = 5$ , how large can the size of  $D$  be?
- 1.33 A digraph  $D$  has vertex set  $\{-3, 3, 6, 12\}$  and  $(i, j) \in D$  if  $i \neq j$  and  $i \mid j$ . Draw the digraph  $D$ .



## Chapter 2

# Degrees

### 2.1 The Degree of a Vertex

There are many numbers, referred to as **parameters**, associated with a graph  $G$ . Knowing the values of certain parameters provides us with information about  $G$ , but rarely tells us the entire structure of  $G$ . (These comments are tied in with the concept of isomorphic graphs, which will be discussed in Chapter 3.) We've already mentioned the best known parameters: the order and the size. There are also numbers associated with each vertex of a graph. We now consider the best known of these.

The **degree of a vertex**  $v$  in a graph  $G$  is the number of edges incident with  $v$  and is denoted by  $\deg_G v$ , or simply by  $\deg v$  if the graph  $G$  is clear from the context. Also,  $\deg v$  is the number of vertices adjacent to  $v$ . Two adjacent vertices are referred to as **neighbors** of each other. The set  $N(v)$  of neighbors of a vertex  $v$  is called the **neighborhood** of  $v$ . Thus  $\deg v = |N(v)|$ .

A vertex of degree 0 is referred to as an **isolated vertex** and a vertex of degree 1 is an **end-vertex** (or a **leaf**). The **minimum degree** of  $G$  is the minimum degree among the vertices of  $G$  and is denoted by  $\delta(G)$ ; the **maximum degree** of  $G$  is denoted by  $\Delta(G)$ . So if  $G$  is a graph of order  $n$  and  $v$  is any vertex of  $G$ , then

$$0 \leq \delta(G) \leq \deg v \leq \Delta(G) \leq n - 1.$$

The graph  $G$  of Figure 2.1 has order 6 and size 5. Each vertex of  $G$  is labeled with its degree. Since  $G$  contains an isolated vertex, namely  $u$ , it follows that  $\delta(G) = 0$ . Furthermore,  $w$  has the largest degree in  $G$  and so  $\Delta(G) = \deg w = 4$ . Both  $v$  and  $z$  are end-vertices of  $G$  since  $\deg v = \deg z = 1$ . If we add the degrees of the vertices of  $G$ , we obtain  $0 + 1 + 1 + 2 + 2 + 4 = 10$ , which happens to be twice the size of  $G$ . This is not a coincidence as we show in our next theorem, which is often referred to as “**The First Theorem of Graph Theory**”, so-called because it is likely that anyone studying graph theory for the first time would

discover this result as his or her own first theorem on the subject. Although we've already discovered some theorems in Chapter 1, we'll follow the trend and also refer to the following theorem as the First Theorem of Graph Theory.

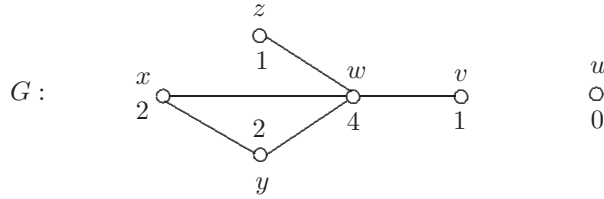


Figure 2.1: A graph  $G$  with  $\delta(G) = 0$  and  $\Delta(G) = 4$

**Theorem 2.1 (The First Theorem of Graph Theory)** *If  $G$  is a graph of size  $m$ , then*

$$\sum_{v \in V(G)} \deg v = 2m.$$

**Proof.** [direct proof] When summing the degrees of the vertices of  $G$ , we count each edge  $e$  of  $G$  twice, once for each of the two vertices incident with  $e$ . ■

The First Theorem of Graph Theory is useful in solving the following problem.

**Example 2.2** *A certain graph  $G$  has order 14 and size 27. The degree of each vertex of  $G$  is 3, 4, or 5. There are six vertices of degree 4. How many vertices of  $G$  have degree 3 and how many have degree 5?*

**Solution.** Let  $x$  be the number of vertices of  $G$  having degree 3. Since the order of  $G$  is 14 and six vertices have degree 4, eight vertices have degree 3 or 5. So there are  $8 - x$  vertices of degree 5. Summing the degrees of the vertices of  $G$  and applying the First Theorem of Graph Theory, we obtain

$$\begin{aligned} 3 \cdot x + 4 \cdot 6 + 5 \cdot (8 - x) &= 2 \cdot 27 \\ 3x + 24 + 40 - 5x &= 54 \\ -2x &= -10 \\ x &= 5 \end{aligned}$$

and so  $8 - x = 3$ . Thus  $G$  has five vertices of degree 3 and three vertices of degree 5. ◇

The method we used to solve the problem in Example 2.2 tells us that there is a unique solution. Perhaps other methods of solving this problem might have occurred to you, such as trying to draw the graph. Consider the graph of Figure 2.2, each of whose vertices is labeled by its degree. This graph has



order 14, size 27, and six vertices of degree 4, which are characteristics of the graph  $G$  of Example 2.2. We see that the graph of Figure 2.2 has five vertices of degree 3 and three vertices of degree 5, solving the problem. Even though this provides the correct answers to our question, the explanation is not correct; for how do we know that the graph we have just drawn is *the* graph  $G$  referred to in the problem and therefore gives us the correct answer?

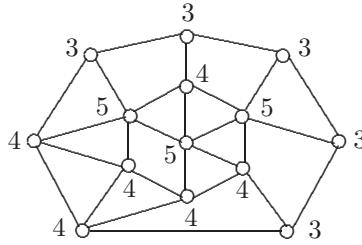


Figure 2.2: A graph of order 14 and size 27

Another possible “solution” might go something like this: We know that we are looking for integers  $x$  and  $y$  such that  $x + y = 8$  and

$$3x + 4 \cdot 6 + 5y = 2 \cdot 27 = 54.$$

By observation, we see that  $x = 5$  and  $y = 3$  satisfy this equation. Thus we have “solved” the problem. But this “solution” also has its drawbacks. How do we know that this is the *only* solution? (Of course, we could try all possible values of  $x$  and  $y$ .) The solution that we gave for Example 2.2 shows that there is only one solution for each of  $x$  and  $y$  and that the solutions do not depend on the graph under consideration (provided it has order 14, size 27, and six vertices of degree 4). Just as when asked to solve  $x^2 - x = 3x - 4$  for a real number  $x$ , it is not enough to simply note that  $x = 2$  is a root. It is required to find *all* roots, and even if  $x = 2$  is the *only* root, we are obliged to show that this is so.

Suppose that  $G$  is a bipartite graph of size  $m$  with partite sets  $U = \{u_1, u_2, \dots, u_s\}$  and  $W = \{w_1, w_2, \dots, w_t\}$ . Since every edge of  $G$  joins a vertex of  $U$  and a vertex of  $W$ , it follows that adding the degrees of the vertices in  $U$  (or in  $W$ ) gives the number of edges in  $G$ , that is,

$$\sum_{i=1}^s \deg u_i = \sum_{j=1}^t \deg w_j = m. \quad (2.1)$$

A vertex of even degree is called an **even vertex**, while a vertex of odd degree is an **odd vertex**. Returning to the graph  $G$  of Figure 2.2, we see that it has six even vertices and eight odd vertices. In particular, the number of odd vertices of  $G$  is even. We show that this is the case for every graph.

**Corollary 2.3** *Every graph has an even number of odd vertices.*

**Proof.** [direct proof] Let  $G$  be a graph of size  $m$ . Divide  $V(G)$  into two subsets  $V_1$  and  $V_2$ , where  $V_1$  consists of the odd vertices of  $G$  and  $V_2$  consists of the even vertices of  $G$ . By the First Theorem of Graph Theory,

$$\sum_{v \in V(G)} \deg v = \sum_{v \in V_1} \deg v + \sum_{v \in V_2} \deg v = 2m.$$

The number  $\sum_{v \in V_2} \deg v$  is even since it is a sum of even integers. Thus

$$\sum_{v \in V_1} \deg v = 2m - \sum_{v \in V_2} \deg v,$$

which implies that  $\sum_{v \in V_1} \deg v$  is even. Since each of the numbers  $\deg v$ ,  $v \in V_1$ , is odd, the number of odd vertices of  $G$  is even. ■

There is a great deal of information that can be learned about a graph from the degrees of its vertices. For example, if a graph  $G$  of order  $n$  contains a vertex of degree  $n - 1$ , then  $G$  is connected. In order to see why this is true, suppose that  $\deg w = n - 1$ . Therefore,  $w$  is adjacent to all other vertices of  $G$ . To show that  $G$  is connected, we need to show that every pair  $x, y$  of vertices of  $G$  are connected, that is,  $G$  contains an  $x - y$  path. This is certainly true if one of  $x$  and  $y$  is  $w$ . If neither  $x$  nor  $y$  is  $w$ , then since  $w$  is adjacent to both  $x$  and  $y$ , it follows that  $x, w, y$  is an  $x - y$  path and consequently  $G$  contains an  $x - y$  path.

This degree condition is certainly not necessary for a graph to be connected. For example, for  $n \geq 4$ , the path  $P_n$  of order  $n$  is connected but contains no vertex of degree greater than 2. Next, we present another degree condition that implies that a graph is connected, and more.

**Theorem 2.4** *Let  $G$  be a graph of order  $n$ . If*

$$\deg u + \deg v \geq n - 1$$

*for every two nonadjacent vertices  $u$  and  $v$  of  $G$ , then  $G$  is connected and  $\text{diam}(G) \leq 2$ .*

**Proof.** [direct proof] We show that every two distinct vertices of  $G$  are connected by a path of length at most 2. Let  $x, y \in V(G)$ . If  $xy \in E(G)$ , then  $x, y$  is a path and  $x$  and  $y$  are certainly connected. Hence we may assume that  $xy \notin E(G)$ . Therefore,  $\deg x + \deg y \geq n - 1$ , which implies that there must be a vertex  $w$  that is adjacent to both  $x$  and  $y$ . So  $x, w, y$  is a path in  $G$ , as desired. ■

Theorem 2.4 implies that if  $G$  is a graph of order  $n$  such that  $\deg v \geq (n - 1)/2$  for every vertex  $v$  of  $G$ , then  $G$  must be connected.

**Corollary 2.5** *If  $G$  is a graph of order  $n$  with  $\delta(G) \geq (n - 1)/2$ , then  $G$  is connected.*

**Proof.** [direct proof] For every two nonadjacent vertices  $u$  and  $v$  of  $G$ ,

$$\deg u + \deg v \geq \frac{n-1}{2} + \frac{n-1}{2} = n-1.$$

By Theorem 2.4,  $G$  is connected. ■

According to Corollary 2.5, if  $G$  is a graph of order  $n = 7$  and  $\delta(G) \geq (7-1)/2 = 3$ , then  $G$  is connected. Also, if  $G$  is a graph of order  $n = 8$  and  $\delta(G) \geq (8-1)/2 = 3.5$ , then  $G$  is connected. Of course, in the latter case, this says that if  $G$  is a graph of order  $n = 8$  and  $\delta(G) \geq 4$ , then  $G$  is connected. For an even integer  $n$ , Corollary 2.5 then says that if  $G$  is a graph of order  $n$  with  $\delta(G) \geq n/2$ , then  $G$  is connected.

Let's return to Theorem 2.4. This theorem tells us then that if the sum of the degrees of any two nonadjacent vertices of a graph  $G$  of order  $n$  is “large enough”, then  $G$  is connected. According to Theorem 2.4,  $n-1$  is large enough. Obviously, if the sum of the degrees of any two nonadjacent vertices of  $G$  is at least  $n$ , then  $G$  must be connected as well. But what if the sum of the degrees of any two nonadjacent vertices of  $G$  is at least  $n-2$ ? Does that also guarantee that  $G$  is connected? What we are now discussing is the **sharpness** of the bound in Theorem 2.4. That is, would Theorem 2.4 still be true if we replace  $n-1$  by a smaller integer? If not, then Theorem 2.4 cannot be improved and the bound is sharp.

As it turns out, the bound in Theorem 2.4 *is* sharp. For example, suppose that  $n$  is even, say  $n = 2k$ , and consider the graph  $G \cong 2K_k$ ; that is,  $G$  is the disconnected graph with two components, each of which is  $K_k$  (see Figure 2.3). If  $u$  and  $v$  are two nonadjacent vertices in this graph  $G$ , then  $u$  and  $v$  must be in different components and each has degree  $k-1$ . So

$$\deg u + \deg v = (k-1) + (k-1) = 2k-2 = n-2.$$

Therefore, if the sum of the degrees of any two nonadjacent vertices of a graph  $G$  of order  $n$  is at least  $n-2$ , then there is no guarantee that  $G$  is connected.

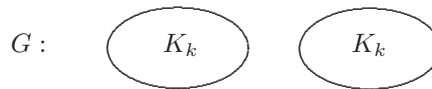


Figure 2.3: A disconnected graph of order  $n = 2k$  such that the sum of the degrees of any two nonadjacent vertices is  $n-2$

Observe also that if  $G$  is a disconnected graph of order  $n$ , then (since  $G$  has at least two components) some component  $G_1$  of  $G$  has order  $n_1$  that is at most  $n/2$ . Every vertex of  $G_1$  has degree at most  $n_1-1 \leq (n/2)-1 = (n-2)/2$  and so  $\delta(G) \leq (n-2)/2$ . (This observation actually provides a proof by contrapositive of Corollary 2.5.) If  $G$  has three components, then the order of some component of  $G$  is at most  $n/3$ . More generally, if  $G$  has  $k$  components, then the order of some component of  $G$  is at most  $n/k$ .

The concept of degree has counterparts in both multigraphs and digraphs. For a vertex  $v$  in a multigraph  $G$ , the **degree**  $\deg v$  of  $v$  in  $G$  is the number of edges of  $G$  incident with  $v$ , where there is a contribution of 2 for each loop at  $v$ . For the multigraph  $G$  of Figure 2.4,

$$\deg u_1 = 4, \deg u_2 = 6, \deg u_3 = 6, \deg u_4 = 4.$$

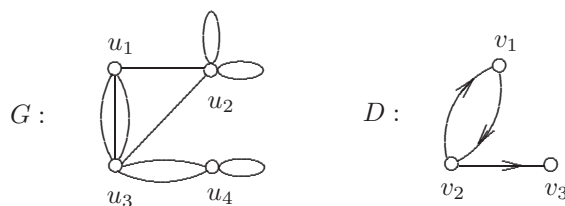


Figure 2.4: Illustrating degrees in a multigraph and a digraph

For a vertex  $v$  in a digraph  $D$ , the **outdegree**  $\text{od } v$  of  $v$  is the number of vertices of  $D$  to which  $v$  is adjacent, while the **indegree**  $\text{id } v$  of  $v$  is the number of vertices of  $D$  from which  $v$  is adjacent. For the digraph  $D$  of Figure 2.4,

$$\text{od } v_1 = \text{id } v_1 = 1, \text{od } v_2 = 2, \text{id } v_2 = 1, \text{od } v_3 = 0, \text{id } v_3 = 1.$$

### Exercises for Section 2.1

2.1 Give an example of the following or explain why no such example exists:

- (a) a graph of order 7 whose vertices have degrees 1, 1, 1, 2, 2, 3, 3.
- (b) a graph of order 7 whose vertices have degrees 1, 2, 2, 2, 3, 3, 7.
- (c) a graph of order 4 whose vertices have degrees 1, 3, 3, 3.

2.2 Give an example of the following or explain why no such example exists:

- (a) a graph that has no odd vertices.
- (b) a noncomplete graph, all of whose vertices have degree 3.
- (c) a graph  $G$  of order 5 or more with the property that  $\deg u \neq \deg v$  for every pair  $u, v$  of adjacent vertices of  $G$ .
- (d) a noncomplete graph  $H$  of order 5 or more with the property that  $\deg u \neq \deg v$  for every pair  $u, v$  of nonadjacent vertices of  $H$ .

2.3 The degree of each vertex of a certain graph of order 12 and size 31 is either 4 or 6. How many vertices of degree 4 are there?

2.4 Give an example of a graph  $G$  of order 6 and size 10 such that  $\delta(G) = 3$  and  $\Delta(G) = 4$ .

- 2.5 The degree of every vertex of a graph  $G$  of order 25 and size 62 is 3, 4, 5, or 6. There are two vertices of degree 4 and 11 vertices of degree 6. How many vertices of  $G$  have degree 5?
- 2.6 Prove that if a graph of order  $3n$  ( $n \geq 1$ ) has  $n$  vertices of each of the degrees  $n-1$ ,  $n$ , and  $n+1$ , then  $n$  is even.
- 2.7 Let  $G$  be a bipartite graph of order 22 with partite sets  $U$  and  $W$ , where  $|U| = 12$ . Suppose that every vertex in  $U$  has degree 3; while every vertex of  $W$  has degree 2 or 4. How many vertices of  $G$  have degree 2?
- 2.8 Let  $G$  be a graph of order  $n$ . If  $\deg u + \deg v + \deg w \geq n-1$  for every three pairwise nonadjacent vertices  $u, v$ , and  $w$  of  $G$ , must  $G$  be connected?
- 2.9 Show that if  $G$  is a disconnected graph containing exactly two odd vertices, then these odd vertices must be in the same component of  $G$ .
- 2.10 We have already seen that if  $G$  is a graph of order  $n$  such that  $\deg u + \deg v \geq n-2$  for every two nonadjacent vertices  $u$  and  $v$  of  $G$ , then  $G$  might be disconnected.
- (a) Show that there exists a connected graph  $G$  of order  $n$  such that  $\deg u + \deg v \geq n-2$  for every two nonadjacent vertices  $u$  and  $v$  and for which  $\deg x + \deg y = n-2$  for some pair  $x, y$  of nonadjacent vertices of  $G$ .
  - (b) Let  $G$  be a graph of order  $n$ . Prove that if  $\deg u + \deg v \geq n-2$  for every pair  $u, v$  of nonadjacent vertices of  $G$ , then  $G$  has at most two components.
  - (c) Is the bound in part (b) sharp?
- 2.11 Corollary 2.5 states that if  $G$  is a graph of order  $n$  with  $\delta(G) \geq (n-1)/2$ , then  $G$  is connected. Is the bound  $(n-1)/2$  sharp, that is, in this case, can  $(n-1)/2$  be replaced by  $(n-2)/2$ ?
- 2.12 Prove that if  $G$  is a graph of order  $n$  such that  $\Delta(G) + \delta(G) \geq n-1$ , then  $G$  is connected and  $\text{diam}(G) \leq 4$ . Show that the bound  $n-1$  is sharp.
- 2.13 Let  $G$  be a graph of order  $n \geq 2$ .
- (a) Prove that if  $\deg v \geq (n-2)/3$  for every vertex  $v$  of  $G$ , then  $G$  contains at most two components.
  - (b) Show that the bound in (a) is sharp.
- 2.14 A graph  $G$  has the property that every edge of  $G$  joins an odd vertex with an even vertex. Show that  $G$  is bipartite and has even size.
- 2.15 A certain connected graph  $G$  has the property that for every two vertices  $u$  and  $v$  of  $G$ , the length of each  $u-v$  path is even or the length of each  $u-v$  path is odd. Prove that  $G$  is bipartite.

- 2.16 The degree of every vertex of a graph  $G$  of order  $2n + 1 \geq 5$  is either  $n + 1$  or  $n + 2$ . Prove that  $G$  contains at least  $n + 1$  vertices of degree  $n + 2$  or at least  $n + 2$  vertices of degree  $n + 1$ .
- 2.17 Let  $G$  be a connected graph containing a vertex  $w$  such that  $\deg w \not\equiv 0 \pmod{3}$  and such that  $\deg u + \deg v \equiv 0 \pmod{3}$  for every two adjacent vertices  $u$  and  $v$  of  $G$ . Prove that  $G$  is bipartite and contains no vertex  $x$  such that  $\deg x \equiv 0 \pmod{3}$ .
- 2.18 Let  $G$  be a graph of order 8 with  $V(G) = \{v_1, v_2, \dots, v_8\}$  such that  $\deg v_i = i$  for  $1 \leq i \leq 7$ . What is  $\deg v_8$ ? Explain.

## 2.2 Regular Graphs

We have already mentioned that  $0 \leq \delta(G) \leq \Delta(G) \leq n - 1$  for every graph  $G$  of order  $n$ . If  $\delta(G) = \Delta(G)$ , then the vertices of  $G$  have the same degree and  $G$  is called **regular**. If  $\deg v = r$  for every vertex  $v$  of  $G$ , where  $0 \leq r \leq n - 1$ , then  $G$  is  **$r$ -regular** or **regular of degree  $r$** . The only regular graphs of order 4 or 5 are shown in Figure 2.5. There is no 1-regular or 3-regular graph of order 5, as no graph contains an odd number of odd vertices by Corollary 2.3.

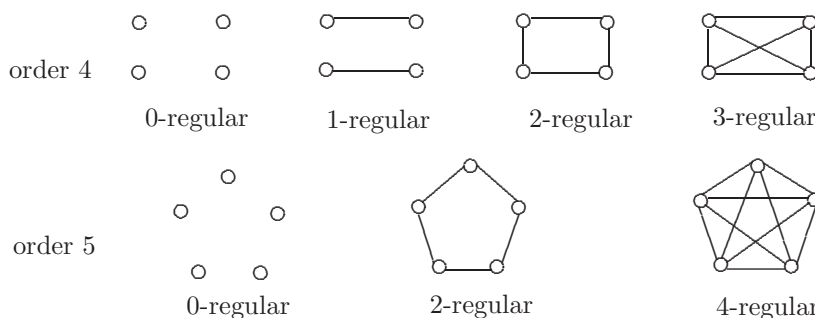


Figure 2.5: Some regular graphs

A 3-regular graph is also referred to as a **cubic graph**. The graphs  $K_4$ ,  $K_{3,3}$ , and  $Q_3$  are cubic graphs; however, the best known cubic graph may very well be the **Petersen graph**, shown in Figure 2.6. We will see this graph again. (Indeed, Section 8.5 is devoted to this graph.)

By Corollary 2.3, there are no  $r$ -regular graphs of order  $n$  if  $r$  and  $n$  are both odd. However, provided  $0 \leq r \leq n - 1$ , there are no other restrictions on the existence of an  $r$ -regular graph of order  $n$ . In the next proof, we will be considering a graph  $G$  with vertex set  $V(G) = \{v_1, v_2, \dots, v_n\}$  and performing

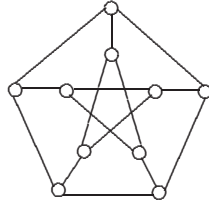


Figure 2.6: The Petersen graph

arithmetic on the subscripts of the vertices. We follow the standard practice of performing the arithmetic modulo  $n$ . For example, if  $n = 6$  and  $i = 5$ , then the vertex  $v_{i+2}$  denotes  $v_1$ .

**Theorem 2.6** *Let  $r$  and  $n$  be integers with  $0 \leq r \leq n - 1$ . There exists an  $r$ -regular graph of order  $n$  if and only if at least one of  $r$  and  $n$  is even.*

**Proof.** [direct proof] As we already mentioned, there is no  $r$ -regular graph of order  $n$  if  $r$  and  $n$  are both odd. It remains only to verify the converse. So let  $r$  and  $n$  be integers with  $0 \leq r \leq n - 1$  such that at least one of  $r$  and  $n$  is even. We construct an  $r$ -regular graph  $H_{r,n}$  of order  $n$ . Let  $V(H_{r,n}) = \{v_1, v_2, \dots, v_n\}$ . First, assume that  $r$  is even. Then  $r = 2k \leq n - 1$  for some nonnegative integer  $k \leq (n - 1)/2$ . For each  $i$  ( $1 \leq i \leq n$ ), we join  $v_i$  to  $v_{i+1}, v_{i+2}, \dots, v_{i+k}$  and to  $v_{i-1}, v_{i-2}, \dots, v_{i-k}$ . If we think of arranging the vertices  $v_1, v_2, \dots, v_n$  cyclically, then each vertex  $v_i$  is adjacent to the  $k$  vertices that immediately follow  $v_i$  and the  $k$  vertices that immediately precede  $v_i$ . Thus  $H_{r,n}$  is  $r$ -regular. For  $r = 4$  and  $n = 10$ , the graph  $H_{4,10}$  is shown in Figure 2.7(a).

Second, assume that  $r$  is odd. Then  $n = 2\ell$  is even. Also,  $r = 2k + 1 \leq n - 1$  for some nonnegative integer  $k \leq (n - 2)/2$ . We join  $v_i$  to the  $2k$  vertices described above as well as to  $v_{i+\ell}$ . In this case, we again think of arranging the vertices  $v_1, v_2, \dots, v_n$  cyclically and joining each vertex  $v_i$  to the  $k$  vertices immediately following it, the  $k$  vertices immediately preceding it, and the unique vertex “opposite”  $v_i$ . Thus  $H_{r,n}$  is  $r$ -regular. For  $r = 5$  and  $n = 10$ , the graph  $H_{5,10}$  is shown in Figure 2.7(b). ■

The graphs  $H_{r,n}$  described above are called **Harary graphs**, named for Frank Harary. We will visit him again. Also, we will visit these graphs again in Section 5.3.

The proof of Theorem 2.6 that we have presented is a *constructive proof*, that is, we actually constructed a graph with the desired properties and didn’t simply show that *some* graph with these properties exists. Although the proof of Theorem 2.6 doesn’t suggest it, there is little restriction on the subgraphs that regular graphs can contain. Of course, if  $H$  is an  $r$ -regular graph, then  $H$  cannot contain any graph  $G$  as a subgraph if  $\Delta(G) > r$ . On the other hand, if  $G$  is a graph with  $\Delta(G) \leq r$  for some integer  $r$ , then  $G$  is a subgraph (indeed, an *induced* subgraph) of some  $r$ -regular graph  $H$ , as we now see.

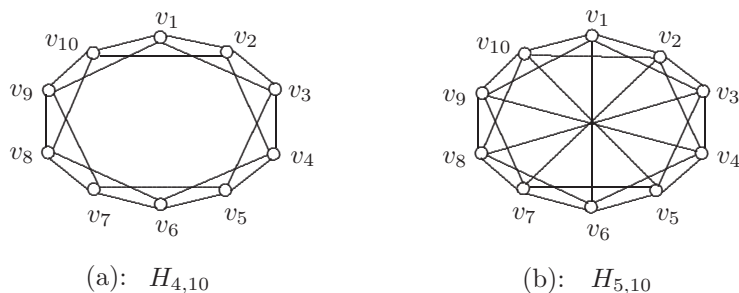


Figure 2.7: A 4-regular graph and a 5-regular graph, both of order 10

**Theorem 2.7** *For every graph  $G$  and every integer  $r \geq \Delta(G)$ , there exists an  $r$ -regular graph  $H$  containing  $G$  as an induced subgraph.*

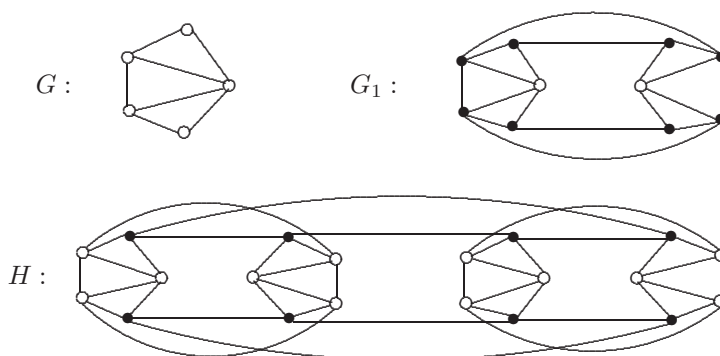
**Proof.** [direct proof] If  $G$  is  $r$ -regular, then we let  $H = G$ . Thus, we may assume that  $G$  is not an  $r$ -regular graph. Suppose that  $G$  has order  $n$  and  $V(G) = \{v_1, v_2, \dots, v_n\}$ . Let  $G'$  be another copy of  $G$  with  $V(G') = \{v'_1, v'_2, \dots, v'_n\}$ , where each vertex  $v'_i$  in  $G'$  corresponds to  $v_i$  in  $G$  for  $1 \leq i \leq n$ . We now construct a graph  $G_1$  from  $G$  and  $G'$  by adding the edges  $v_i v'_i$  for all vertices  $v_i$  ( $1 \leq i \leq n$ ) of  $G$  for which  $\deg v_i < r$ . Then  $G$  is an induced subgraph of  $G_1$  and  $\delta(G_1) = \delta(G) + 1$ . If  $G_1$  is  $r$ -regular, then we let  $H = G_1$ . If not, then we continue this procedure until we arrive at an  $r$ -regular graph  $G_k$ , where  $k = r - \delta(G)$ . The graph  $G_k$  is the desired graph  $H$ . ■

To illustrate the construction described in Theorem 2.7, consider the graph  $G$  of Figure 2.8, where  $\Delta(G) = 4$  and  $\delta(G) = 2$ . We seek a 4-regular graph  $H$  containing  $G$  as an induced subgraph. First, we construct a graph  $G_1$  from two copies of  $G$  by joining all pairs of corresponding vertices in these copies whose degrees are less than 4. Then  $\Delta(G_1) = 4$ ,  $\delta(G_1) = 3$ , and  $G$  is an induced subgraph of  $G_1$ . We then construct  $H$  from two copies of  $G_1$  by joining all pairs of corresponding vertices in these copies whose degrees are 3. Then  $H$  is 4-regular and  $G$  is an induced subgraph of  $H$ .

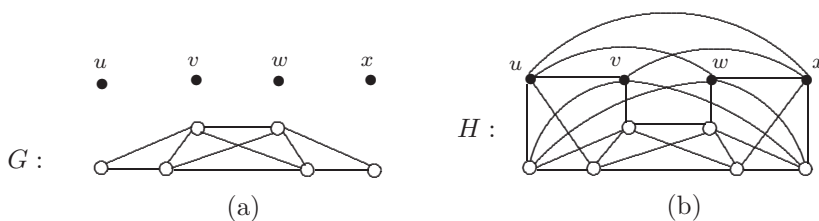
Theorem 2.7 appears in the first book entirely devoted to graph theory, published in 1936 and written by Denés König. Although König stated and proved the theorem for multigraphs, the proof he presented suggests the proof we gave for graphs in Theorem 2.7. The proof of Theorem 2.7 does not construct an  $r$ -regular graph of smallest order containing a given graph  $G$  with  $\Delta(G) \leq r$  as an induced subgraph. Paul Erdős and Paul J. Kelly discovered a method for finding the smallest order of such an  $r$ -regular graph. We consider an example dealing with this problem.

**Example 2.8** *For the graph  $G$  of Figure 2.9, find a 5-regular graph  $H$  of minimum order containing  $G$  as an induced subgraph.*



Figure 2.8: A 4-regular graph  $H$  containing  $G$  as an induced subgraphFigure 2.9: The graph  $G$  in Example 2.8

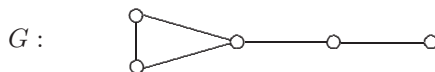
**Solution.** Since  $\delta(G) = 2$  and the order of  $G$  is 6, the order of such a graph  $H$  must be at least 9. However, there does not exist a 5-regular graph of order 9, so the order of  $H$  must be at least 10. Hence to construct a 5-regular graph containing  $G$  as an induced subgraph, it is necessary to add at least four vertices  $u, v, w, x$  to  $G$  (see Figure 2.10(a)). Joining  $u, v, w, x$  to the vertices of  $G$  and to each other as indicated in Figure 2.10(b) produces a 5-regular graph  $H$  containing  $G$  as an induced subgraph. Thus the minimum order of such a graph  $H$  is 10.  $\diamond$

Figure 2.10: The graph  $H$  in Example 2.8

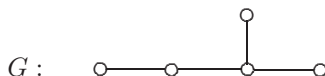
### Exercises for Section 2.2

- 2.19 Construct an  $r$ -regular graph of order 6 and an  $s$ -regular graph of order 7 for all possible values of  $r$  and  $s$ .
- 2.20 Show that if  $G$  is a connected graph that is not regular, then  $G$  contains adjacent vertices  $u$  and  $v$  such that  $\deg u \neq \deg v$ .

- 2.21 (a) Find spanning subgraphs  $G_0, G_1, G_2, G_3$  of the Petersen graph, where  $G_r$  is  $r$ -regular for  $0 \leq r \leq 3$ .
- (b) Find induced subgraphs  $F_0, F_1, F_2, F_3$  in the Petersen graph, where  $F_r$  is  $r$ -regular for  $0 \leq r \leq 3$ .
- (c) How can the problem in (b) be revised so that it would be more interesting (and more challenging)?
- 2.22 For the graph  $G$  of Figure 2.11, construct a 3-regular graph  $H$  containing  $G$  as an induced subgraph
- (a) using the proof of Theorem 2.7. What is the order of  $H$ ?
- (b) such that  $H$  has the smallest possible order. What is this order?

Figure 2.11: The graph  $G$  in Exercise 2.22

- 2.23 Construct a 3-regular graph  $H$  of minimum order containing
- (a)  $P_5$  as an induced subgraph.
- (b)  $P_6$  as an induced subgraph.
- (c)  $P_7$  as an induced subgraph.
- 2.24 What is the minimum order of a 3-regular graph  $H$  containing the graph  $G$  in Figure 2.12 as an induced subgraph?

Figure 2.12: The graph  $G$  in Exercise 2.24

- 2.25 (a) Let  $v$  be a vertex of a graph  $G$ . Show that if  $G - v$  is 3-regular, then  $G$  has odd order.
- (b) Let  $G$  be an  $r$ -regular graph, where  $r$  is odd. Show that  $G$  does not contain any component of odd order.
- 2.26 (a) Show that a graph  $G$  is regular if and only if  $\overline{G}$  is regular.
- (b) Show that if  $G$  and  $\overline{G}$  are both  $r$ -regular for some nonnegative integer  $r$ , then  $G$  has odd order.
- 2.27 Prove that if  $G$  is an  $r$ -regular bipartite graph with  $r \geq 1$  and partite sets  $U$  and  $W$ , then  $|U| = |W|$ .

- 2.28 Investigate the following question: Does there exist a graph  $G$  and an integer  $r$ , where  $\delta(G) < r$  and  $\Delta(G) \leq r$ , such that the  $r$ -regular graph  $H$  in Theorem 2.7 that contains  $G$  as an induced subgraph has smallest order among all  $r$ -regular graphs with this property?
- 2.29 (a) Prove that if  $G$  is a graph of order  $n$ , then  $\delta(G) + \delta(\overline{G}) \leq n - 1$ .  
 (b) Prove, for a graph  $G$  of order  $n$ , that  $\delta(G) + \delta(\overline{G}) = n - 1$  if and only if  $G$  is regular.
- 2.30 Beginning with  $G = K_1$ , use the construction in Theorem 2.7 to produce a 3-regular graph  $H$  (containing  $G$  as an induced subgraph). What famous graph is  $H$ ?

## 2.3 Degree Sequences

Although we've been discussing graphs all of whose vertices have the same degree, it is more typical for the vertices of a graph to have a variety of degrees. If the degrees of the vertices of a graph  $G$  are listed in a sequence  $s$ , then  $s$  is called a **degree sequence** of  $G$ . For example, all of the sequences

$$s : 4, 3, 2, 2, 2, 1, 1, 1, 0; \quad s' : 0, 1, 1, 1, 2, 2, 2, 3, 4; \quad s'' : 4, 3, 2, 1, 2, 2, 1, 1, 0 \quad (2.2)$$

are degree sequences of the graph  $G$  of Figure 2.13, each of whose vertices is labeled by its degree. The sequence  $s$  is non-increasing,  $s'$  is non-decreasing, and  $s''$  is neither. Determining a degree sequence of a graph is not difficult. There is a converse question that is considerably more intriguing, however.

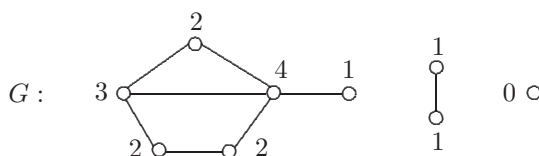


Figure 2.13: A graph with degree sequence  $4, 3, 2, 2, 2, 1, 1, 1, 0$

Suppose that we are given a finite sequence  $s$  of nonnegative integers. Is  $s$  a degree sequence of some graph? A finite sequence of nonnegative integers is called **graphical** if it is a degree sequence of some graph. Of course, all of the sequences in (2.2) are graphical.

**Example 2.9** Which of the following sequences are graphical?

- (1)  $s_1 : 3, 3, 2, 2, 1, 1$

$$(2) \ s_2 : 6, 5, 5, 4, 3, 3, 3, 2, 2$$

$$(3) \ s_3 : 7, 6, 4, 4, 3, 3, 3$$

$$(4) \ s_4 : 3, 3, 3, 1$$

**Solution.**

- (1) The sequence  $s_1$  is graphical. Indeed, it is a degree sequence of the graph  $G_1$  of Figure 2.14.

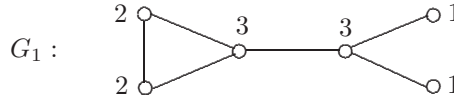


Figure 2.14: A graph with degree sequence 3, 3, 2, 2, 1, 1

- (2) Since  $s_2$  has an odd number of terms that are odd integers,  $s_2$  cannot be a degree sequence of a graph (for otherwise, such a graph would have an odd number of odd vertices, contradicting Corollary 2.3). Therefore,  $s_2$  is not graphical.
- (3) The sequence  $s_3$  is also not graphical; for otherwise,  $s_3$  would be a degree sequence of a graph of order 7 and containing a vertex of degree 7. (The degree of any vertex in a graph of order 7 is at most  $7 - 1 = 6$ .)
- (4) The sequence  $s_4$  contains four terms, all of which are at most  $4 - 1 = 3$ . Also,  $s_4$  contains an even number of terms that are odd integers. Yet,  $s_4$  is not graphical. Assume, to the contrary, that  $s_4$  is graphical. Then there is a graph  $G_4$  of order 4 with  $V(G_4) = \{u, v, w, x\}$  such that  $\deg u = \deg v = \deg w = 3$  and  $\deg x = 1$ . This implies that each of  $u, v$ , and  $w$  is adjacent to all other vertices of  $G_4$ , including  $x$ , but  $x$  is adjacent to only one of  $u, v$ , and  $w$ . This is impossible.  $\diamond$

The sequence  $s_4$  shows that determining which sequences are graphical is potentially difficult. We present a theorem that will help us to efficiently decide whether a given sequence is graphical. This theorem is due to Václav Havel and S. Louis Hakimi and is often referred to as the Havel-Hakimi Theorem, despite the fact that Havel and Hakimi gave independent proofs and wrote separate papers that include this theorem. To use this theorem, we assume that we are beginning with a non-increasing sequence.

**Theorem 2.10** *A non-increasing sequence  $s : d_1, d_2, \dots, d_n$  ( $n \geq 2$ ) of non-negative integers, where  $d_1 \geq 1$ , is graphical if and only if the sequence*

$$s_1 : d_2 - 1, d_3 - 1, \dots, d_{d_1+1} - 1, d_{d_1+2}, \dots, d_n$$

*is graphical.*

**Proof.** [direct proof, proof by contradiction] First, assume that  $s_1$  is graphical. Then there is a graph  $G_1$  with  $V(G_1) = \{v_2, v_3, \dots, v_n\}$  such that

$$\deg_{G_1} v_i = \begin{cases} d_i - 1 & \text{if } 2 \leq i \leq d_1 + 1 \\ d_i & \text{if } d_1 + 2 \leq i \leq n. \end{cases}$$

We construct a graph  $G$  from  $G_1$  by adding a new vertex  $v_1$  and the  $d_1$  edges  $v_1 v_i$  for  $2 \leq i \leq d_1 + 1$ . Since  $\deg_G v_i = d_i$  for  $1 \leq i \leq n$ , it follows that  $s$  is a degree sequence of  $G$  and so  $s$  is graphical.

Proving the converse is more challenging. Assume that  $s$  is graphical. Suppose that a graph  $G$  has degree sequence  $s$  and contains a vertex  $u$  of degree  $d_1$  such that  $u$  is adjacent to vertices whose degrees are  $d_2, d_3, \dots, d_{d_1+1}$ . Then  $s_1$  is a degree sequence of  $H - u$  and the proof of the converse is complete. (However, if no graph with degree sequence  $s$  contains a vertex  $u$  with this property, then what do we do? Fortunately, we need not be concerned with this question as we are about to show that there must be a graph  $H$  with degree sequence  $s$  containing such a vertex  $u$ .)

Assume, to the contrary, that there is no graph with degree sequence  $s$  containing a vertex of degree  $d_1$  that is adjacent to vertices whose degrees are  $d_2, d_3, \dots, d_{d_1+1}$ . Among all graphs with degree sequence  $s$ , let  $G$  be one with  $V(G) = \{v_1, v_2, \dots, v_n\}$  such that  $\deg v_i = d_i$  for  $1 \leq i \leq n$  and the sum of the degrees of vertices adjacent to  $v_1$  is as large as possible. Since  $v_1$  is not adjacent to vertices having degrees  $d_2, d_3, \dots, d_{d_1+1}$  (that is,  $v_1$  is not adjacent to vertices with the next  $d_1$  highest degrees),  $v_1$  must be adjacent to a vertex  $v_s$  having a smaller degree than a vertex  $v_r$  to which  $v_1$  is *not* adjacent. That is, there exist vertices  $v_r$  and  $v_s$  with  $d_r > d_s$  such that  $v_1$  is adjacent to  $v_s$  but not to  $v_r$ . Since  $\deg v_r = d_r > d_s = \deg v_s$ , there exists a vertex  $v_t$  such that  $v_t$  is adjacent to  $v_r$  but not to  $v_s$ . Consider the graph  $G'$  obtained from  $G$  by removing the edges  $v_1 v_s$  and  $v_r v_t$  and adding the edges  $v_1 v_r$  and  $v_s v_t$  (see Figure 2.15, where a dashed line means no edge).

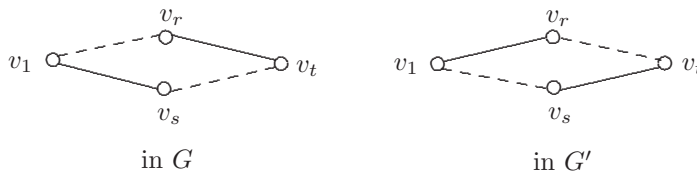


Figure 2.15: Edges in (and not in)  $G$  and  $G'$  in the proof of Theorem 2.10

Then  $G$  and  $G'$  have the same vertex set; indeed,  $s$  is a degree sequence of both  $G$  and  $G'$ . However, the sum of the degrees of the vertices adjacent to  $v_1$  in  $G'$  is larger than that in  $G$ , which produces a contradiction. ■

We now illustrate Theorem 2.10 with the following two examples.

**Example 2.11** *Decide whether the sequence  $s : 5, 4, 3, 3, 2, 2, 2, 1, 1, 1$  is graphical.*

**Solution.** Deleting 5 from  $s$  and subtracting 1 from the next five terms, we obtain

$$s'_1 : 3, 2, 2, 1, 1, 2, 1, 1, 1.$$

Reordering this sequence (so that a non-increasing sequence results and we can apply Theorem 2.10 again), we have

$$s_1 : 3, 2, 2, 2, 1, 1, 1, 1, 1.$$

By Theorem 2.10,  $s$  is graphical if and only if  $s_1$  is. To decide whether  $s_1$  is graphical, we choose to continue this procedure since  $s_1$  is relatively complicated. Deleting 3 from  $s_1$  and subtracting 1 from the next three terms, we obtain

$$s'_2 = s_2 : 1, 1, 1, 1, 1, 1, 1, 1.$$

By Theorem 2.10,  $s_1$  is graphical if and only if  $s_2$  is. But  $s_2$  is so simple that we can quickly observe that  $s_2$  is a (actually *the*) degree sequence of the graph  $G_2 = 4K_2$  of Figure 2.16. Therefore,  $s_2$  is graphical, so  $s_1$  and  $s$  are graphical as well.  $\diamond$

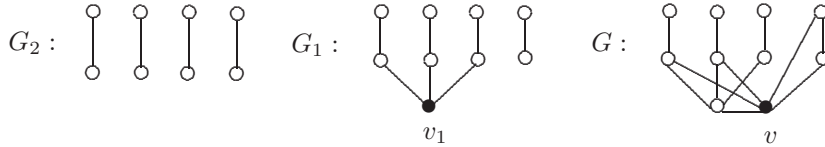


Figure 2.16: The graphs  $G_2$ ,  $G_1$ , and  $G$  in Example 2.11

In Example 2.11, we stopped when we arrived at the sequence  $s_2$ , as it was clear that  $s_2$  is a degree sequence of some graph, namely  $s_2$  is the degree sequence of the graph  $G_2$  of Figure 2.16. Recall that the sequence  $s_2 = s'_2$  was obtained from  $s_1$  by deleting 3 from  $s_1$  and subtracting 1 from the next three terms of  $s_1$ . Consequently, if we add a new vertex  $v_1$  to  $G_2$  and join  $v_1$  to three vertices of degree 1 in  $G_2$ , then we obtain a graph  $G_1$  (see Figure 2.16) with degree sequence  $s_1$ .

In Example 2.11, the sequence  $s_1$  was obtained from  $s'_1$  by rearranging its terms to produce a decreasing sequence. The sequence  $s'_1$  was, in turn, obtained from  $s$  by deleting the first term 5 from  $s$  and subtracting 1 from the next five terms of  $s$ . Therefore, if we add a new vertex  $v$  to  $G_1$  and join  $v$  to the vertex of degree 3, to two vertices of degree 2, and to two vertices of degree 1, then we obtain a graph  $G$  with degree sequence  $s$ . So we have just constructed a graph with degree sequence  $s$ . Informally, we have just described an efficient algorithm for deciding whether a given sequence of nonnegative integers is graphical. We present one additional example to illustrate how to proceed when the sequence is not graphical.

**Example 2.12** *Decide whether the sequence  $s : 7, 7, 4, 3, 3, 3, 2, 1$  is graphical.*

**Solution.** Deleting the first term 7 from  $s$  and subtracting 1 from the next seven terms of  $s$ , we obtain

$$s_1 = s'_1 : 6, 3, 2, 2, 2, 1, 0.$$

Perhaps we can see that  $s_1$  is not graphical, but nevertheless let's continue. Deleting the first term 6 from  $s_1$  and subtracting 1 from the next six terms of  $s_1$ , we obtain

$$s_2 = s'_2 : 2, 1, 1, 1, 0, -1.$$

Since  $s_2$  contains the negative number  $-1$  and, of course, no vertex can have a negative degree, it is now very clear that  $s_2$  is not graphical. By Theorem 2.10, the sequence  $s_1$  is not graphical and so neither is  $s$ .  $\diamond$

Another way to show that the sequence in Example 2.12 is not graphical is to observe that if it were graphical, then there would be a graph  $G$  of order 8 with degree sequence  $s$ , implying that  $G$  has two vertices of degree 7 and so all other vertices of  $G$  have degree at least 2. Hence  $G$  can have no vertex of degree 1.

### Exercises for Section 2.3

- 2.31 Prove that a sequence  $d_1, d_2, \dots, d_n$  is graphical if and only if  $n - d_1 - 1, n - d_2 - 1, \dots, n - d_n - 1$  is graphical.
- 2.32 Use Theorem 2.10 to determine which of the following sequences are graphical. For each of those that are graphical, construct a graph, as in Example 2.11, for which the given sequence is a degree sequence of the graph.
- (a)  $s_1 : 5, 3, 3, 3, 3, 2, 2, 1$
  - (b)  $s_2 : 6, 3, 3, 3, 3, 2, 2, 2, 1, 1$
  - (c)  $s_3 : 6, 5, 5, 4, 3, 2, 1$
  - (d)  $s_4 : 7, 5, 4, 4, 4, 3, 2, 1$
  - (e)  $s_5 : 7, 6, 5, 4, 4, 3, 2, 1$
- 2.33 Prove that for every integer  $x$  with  $0 \leq x \leq 5$ , the sequence  $x, 1, 2, 3, 5, 5$  is not graphical.
- 2.34 For which integers  $x$  ( $0 \leq x \leq 7$ ), if any, is the sequence  $7, 6, 5, 4, 3, 2, 1, x$  graphical?
- 2.35 If the sequence  $x, 7, 7, 5, 5, 4, 3, 2$  is graphical, then what are the possible values of  $x$  ( $0 \leq x \leq 7$ )?
- 2.36 Let  $S = \{2, 6, 7\}$ . Prove that there exists a positive integer  $k$  such that the sequence obtained by listing each element of  $S$  a total  $k$  times is a degree sequence of some graph. What is the minimum value of  $k$ ?

## 2.4 Excursion: Graphs and Matrices

As we know, a graph  $G$  can be defined by two sets, namely its vertex set  $V(G)$  and edge set  $E(G)$ , or by a diagram. A graph can also be described by a matrix and for some purposes this is especially useful.

Let  $G$  be a graph of order  $n$  and size  $m$ , where  $V(G) = \{v_1, v_2, \dots, v_n\}$  and  $E(G) = \{e_1, e_2, \dots, e_m\}$ . The **adjacency matrix** of  $G$  is the  $n \times n$  matrix  $A = [a_{ij}]$ , where

$$a_{ij} = \begin{cases} 1 & \text{if } v_i v_j \in E(G) \\ 0 & \text{otherwise;} \end{cases}$$

while the **incidence matrix** of  $G$  is the  $n \times m$  matrix  $B = [b_{ij}]$ , where

$$b_{ij} = \begin{cases} 1 & \text{if } v_i \text{ is incident with } e_j \\ 0 & \text{otherwise.} \end{cases}$$

These matrices are shown for the graph  $G$  of Figure 2.17.

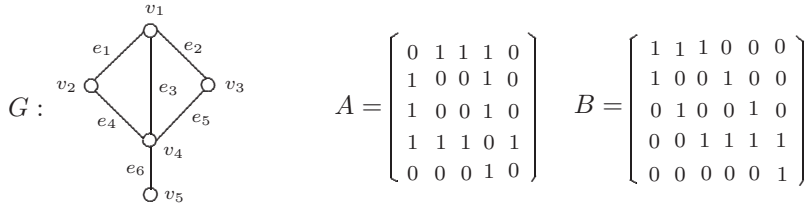


Figure 2.17: The adjacency matrix and incidence matrix of a graph

Here are a few useful observations about the adjacency matrix and incidence matrix. First, these matrices are dependent on how the vertices and edges of  $G$  are labeled. In any case, the adjacency matrix is a symmetric  $n \times n$  matrix where every entry on the main diagonal is 0. The number of 1s in row  $i$  (or column  $i$ ) is the degree of the vertex  $v_i$ . While the number of 1s in row  $i$  of the incidence matrix is also the degree of  $v_i$ , the number of 1s in each of its columns is 2 since there are exactly two vertices incident with every edge.

Two  $u - v$  walks are considered **equal** if, as sequences, they are identical, term by term. Let's now return to the graph  $G$  and its adjacency matrix  $A$  shown in Figure 2.17. The square  $A^2$  and the cube  $A^3$  of  $A$  are given in Figure 2.18. If we look at the entries along the main diagonal in  $A^2$ , we see that these are the degrees of the vertices of  $G$ . This is not only true in general, but each entry in each power of  $A$  represents some characteristic of the graph  $G$ .

**Theorem 2.13** *Let  $G$  be a graph with vertex set  $V(G) = \{v_1, v_2, \dots, v_n\}$  and adjacency matrix  $A = [a_{ij}]$ . Then the entry  $a_{ij}^{(k)}$  in row  $i$  and column  $j$  of  $A^k$  is the number of distinct  $v_i - v_j$  walks of length  $k$  in  $G$ .*



$$A^2 = \begin{bmatrix} 3 & 1 & 1 & 2 & 1 \\ 1 & 2 & 2 & 1 & 1 \\ 1 & 2 & 2 & 1 & 1 \\ 2 & 1 & 1 & 4 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{bmatrix} \quad A^3 = \begin{bmatrix} 4 & 5 & 5 & 6 & 2 \\ 5 & 2 & 2 & 6 & 1 \\ 5 & 2 & 2 & 6 & 1 \\ 6 & 6 & 6 & 4 & 4 \\ 2 & 1 & 1 & 4 & 0 \end{bmatrix}$$

Figure 2.18: Powers of an adjacency matrix

**Proof.** [induction] We proceed by induction. We begin with  $A = A^1$ . Each entry  $a_{ij} = a_{ij}^{(1)}$  of  $A$  is either 1 or 0, according to whether  $v_i v_j$  is or is not an edge of  $G$ . Hence  $a_{ij}^{(1)}$  gives the number of  $v_i - v_j$  walks of length 1 in  $G$ . Assume, for a positive integer  $k$ , that the number of  $v_i - v_j$  walks of length  $k$  in  $G$  is given by  $a_{ij}^{(k)}$ . From the definition of matrix multiplication, the  $(i, j)$ -entry  $a_{ij}^{(k+1)}$  in  $A^{k+1}$  is the dot product of row  $i$  of  $A^k$  and column  $j$  of  $A$ , that is,

$$a_{ij}^{(k+1)} = \sum_{t=1}^n a_{it}^{(k)} a_{tj} = a_{i1}^{(k)} a_{1j} + a_{i2}^{(k)} a_{2j} + \dots + a_{in}^{(k)} a_{nj}. \quad (2.3)$$

Every  $v_i - v_j$  walk  $W$  of length  $k + 1$  is produced by beginning with a  $v_i - v_t$  walk  $W'$  of length  $k$  for some vertex  $v_t$  adjacent to  $v_j$  and then following  $W'$  by  $v_j$ . By the induction hypothesis, the number of  $v_i - v_t$  walks of length  $k$  is  $a_{it}^{(k)}$ , while  $v_t$  is adjacent to  $v_j$  if and only if  $a_{tj} = 1$ . Hence by (2.3),  $a_{ij}^{(k+1)}$  does indeed provide the number of  $v_i - v_j$  walks of length  $k + 1$  in  $G$ . ■

In view of Theorem 2.13,  $a_{ii}^{(2)} = \deg v_i$  and  $a_{ii}^{(3)}$  is twice the number of triangles in  $G$  that contain  $v_i$ . Indeed, knowing what each entry of  $A^k$  represents allows us to compute powers of adjacency matrices, at least small powers of adjacency matrices of graphs of small orders without actually multiplying matrices. For example, for the graph  $G$  of Figure 2.17, the entry  $a_{41}^{(3)}$  of  $A^3$  is the number of distinct  $v_4 - v_1$  walks in  $G$  of length 3. Since there are six  $v_4 - v_1$  walks of length 3 in  $G$ , namely:

$$\begin{array}{lll} (1) & v_4, v_5, v_4, v_1 & (2) & v_4, v_2, v_4, v_1 & (3) & v_4, v_3, v_4, v_1 \\ (4) & v_4, v_1, v_2, v_1 & (5) & v_4, v_1, v_3, v_1 & (6) & v_4, v_1, v_4, v_1, \end{array}$$

it follows that  $a_{41}^{(3)} = 6$ , as we have seen.

#### Exercises for Section 2.4

- 2.37 For the adjacency matrix  $A$  of the graph  $G_1$  of Figure 2.19, determine  $A^2$  and  $A^3$  without computing  $A$  or performing matrix multiplication.
- 2.38 For the adjacency matrix  $A$  of the graph  $G_2$  of Figure 2.19, determine  $A^2$  and  $A^3$  without computing  $A$  or performing matrix multiplication.

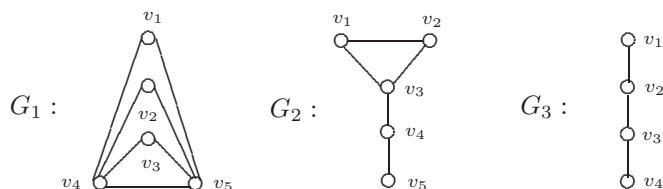


Figure 2.19: Graphs for Exercises 2.37-2.39

- 2.39 For the adjacency matrix  $A$  of the graph  $G_3$  of Figure 2.19, determine  $A^4$  without computing  $A$  or performing matrix multiplication.
- 2.40 For  $G \cong K_{r,r}$  with partite sets  $U = \{v_1, v_2, \dots, v_r\}$  and  $W = \{v_{r+1}, v_{r+2}, \dots, v_{2r}\}$ , determine the adjacency matrix  $A$  of  $G$  and its powers  $A^2$ ,  $A^3$ , and  $A^4$  without performing matrix multiplication.
- 2.41 (a) For the incidence matrix  $B$  in Figure 2.17, compute  $BB^t$ , where  $B^t$  is the transpose of  $B$ .
- (b) For a graph  $G$  with  $V(G) = \{v_1, v_2, \dots, v_n\}$ , what does the  $(i, j)$ -entry of  $BB^t$  represent in  $G$ ?

## 2.5 Exploration: Irregular Graphs

Recall that a graph  $G$  is regular if every two vertices of  $G$  have the same degree. We've already discussed the existence of regular graphs. We now consider graphs that are opposite to regular graphs and the existence of such graphs. A graph  $G$  of order at least 2 is **irregular** if every two vertices of  $G$  have distinct degrees. Before proceeding further with this line of discussion, however, let us pause a bit.

In the Fall 1988 issue (Volume 10, No. 4) of the magazine *Mathematical Intelligencer*, the British mathematics educator and writer David Wells asked the readers to evaluate 24 theorems for their beauty. Two years later (Volume 12, No. 3), he reported the results of the responses he received. The theorem that finished first was

$$e^{i\pi} = -1.$$

There was a tie for second place:

$$\text{Euler's Polyhedral Formula: } V - E + F = 2.$$

*There are infinitely many primes.*